

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUBJECT NAME: COMPUTER NETWORKS

SUBJECT CODE: CST52

Network layer – design issues – Routing algorithms - The Optimality Principle - Shortest Path Algorithm – Flooding - Distance Vector Routing - Link State Routing - Hierarchical Routing - Broadcast Routing - Multicast Routing Congestion Control – Approaches - Traffic-Aware Routing - Admission Control - Traffic Throttling - Load Shedding – Internetworking - Tunneling - Internetwork Routing - Packet Fragmentation - IP v4 - IP Addresses – IPv6 - Internet Control Protocols – OSPF - BGP

2 MARKS

1. What is meant by Network layer?

The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). The network layer ensures that each packet gets from its point of origin to its final destination.

2. Mention the various issues in network layers?

These issues include the service provided to the transport layer and the internal design of the network.

1. Store-and-Forward Packet Switching
2. Services Provided to the Transport Layer
3. Implementation of Connectionless Service
4. Implementation of Connection-Oriented Service
5. Comparison of Virtual-Circuit and Datagram Networks

3. What are the Services provided to the Transport Layer?

1. The services should be independent of the router technology.
2. The transport layer should be shielded from the number, type, and topology of the routers present.
3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

4. What is meant by datagram?

IP transports data in packets called datagrams, each of which is transported separately. Datagrams can travel along different routes and can arrive out of sequence or be duplicated. IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination.

5. What is meant by virtual circuit network?

If connection-oriented service is used, a path from the source router all the way to the destination router must be established before any data packets can be sent. This connection is called a **VC (virtual circuit)**, in analogy with the physical circuits set up by the telephone system, and the network is called a **virtual-circuit network**.

A virtual-circuit network is a cross between a circuit-switched network and a datagram network.

6. What is meant by packet switched network?

In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet. Resources are allocated on demand. The allocation is done on a first come, first-served basis. When a switch receives a packet, no matter what is the source or destination, the packet must wait if there are other packets being processed.

7. Comparison between datagram network and virtual network?

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

8. What is routing algorithm?

The routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on. If the network uses datagrams internally, this decision must be made anew for every arriving data packet since the best route may have changed since last time.

9. What is mean by session routing?

Session routing because a route remains in force for an entire session (e.g., while logged in over a VPN).

10. What is forwarding?

- One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. This process is forwarding.
- The other process is responsible for filling in and updating the routing tables.
- Forwarding means to place the packet in its route to its destination.
- Forwarding requires a host or a router to have a routing table.
- When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this table to find the route to the final destination.

11. Mention two major classes in routing algorithm?

Routing algorithms can be grouped into two major classes

- Non-adaptive routing algorithm
- Adaptive routing algorithm

12. What is Adaptive and Non-adaptive routing algorithm?

- Non-adaptive algorithms do not base their routing decisions on any measurements or estimates of the current topology and traffic.
- Adaptive algorithms, in contrast, change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well.

13. What is static and dynamic routing?

- The choice of the route to use to get from I to J is computed in advance, offline, and downloaded to the routers when the network is booted. This procedure is sometimes called static routing.
- The dynamic routing algorithms differ in where they get their information, when they change the routes and what metric is used for optimization.

14. List the variety of routing algorithms?

- The Optimality Principle
- Shortest Path Algorithm
- Flooding
- Distance Vector Routing
- Link State Routing
- Hierarchical Routing
- Broadcast Routing
- Multicast Routing

15. Define sink tree?

The set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a sink tree.

16. Describe the shortest path algorithm?

The idea is to build a graph of the network, with each node of the graph representing a router and each edge of the graph representing a communication line, or link. To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.

17. Define flooding?

- A simple local technique is flooding, in which every incoming packet is sent out on every outgoing line except the one it arrived on.
- Flooding generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the process.

18. What is mean by distance vector routing?

- The distance vector routing algorithm is sometimes called distributed Bellman-Ford routing algorithm.
- A distance vector routing algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there. These tables are updated by exchanging information with the neighbors. Eventually, every router knows the best link to reach each destination.

- In distance vector routing, the least-cost route between any two nodes is the route with minimum distance.
- In this protocol, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).

19. State the purpose of maintaining the routing table and where is it maintained? (MAY 2017)

- In distance vector routing, each router maintains a routing table indexed by, and containing one entry for each router in the network.
- The entry has two parts are
 - The preferred outgoing line to use for that destination and an estimate of the distance to that destination.
 - The distance might be measured as the number of hops or using another metric.

20. What is mean by Link state routing?

- In link state routing, if each node in the domain has the entire topology of the domain the list of nodes and links, how they are connected including the type, cost (metric), and condition of the links (up or down)-the node can use Dijkstra's algorithm to build a routing table.
- Link state routing called IS-IS and OSPF are the routing algorithms that are most widely used inside large networks and the Internet today.

21. Mention the five approaches in Link state routing?

1. Learning about the Neighbors
2. Setting Link Costs
3. Building Link State Packets
4. Distributing the Link State Packets
5. Computing the New Routes

22. Define OSPF?

The Open Shortest Path First or OSPF protocol is an intra-domain routing protocol based on link state routing. Its domain is also an autonomous system.

23. What is mean by Hierarchical Routing?

- A routing technique in which the entire address space is divided into levels based on specific criteria.
- When hierarchical routing is used, the routers are divided into what we will call **regions**.
- Each router knows all the details about how to route packets to destinations within its own region but knows nothing about the internal structure of other regions.

24. What is mean by broadcasting routing?

- Sending a packet to all destinations simultaneously is called broadcasting.
- In broadcast communication, the relationship between the source and the destination is one-to-all. There is only one source, but all the other hosts are the destinations.
- Broadcasting method that requires no special features from the network is for the source to simply send a distinct packet to each destination

25. Define multi-destination routing?

Multi-destination routing, in which each packet contains either a list of destinations or a bit map indicating the desired destinations. When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed.

26. What is reverse path forwarding?

When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets toward the source of the broadcast.

27. What is spanning tree?

A spanning tree is a subset of the network that includes all the routers but contains no loops. Sink trees are spanning trees. If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.

28. Define multicast routing?

Sending a message to such a group is called **multicasting**, and the routing algorithm used is called **multicast routing**. All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group.

29. Define MOSPF?

Multicast Open Shortest Path First (MOSPF) protocol is an extension of the OSPF protocol that uses multicast link state routing to create source-based trees.

30. Define DVMRP?

The Distance Vector Multicast Routing Protocol (DVMRP) is an implementation of multicast distance vector routing. It is a source-based routing protocol, based on RIP.

31. Define CBT?

A Core-Based Trees (CBT) to compute a single spanning tree for the group. All of the routers agree on a root (called the **core** or **rendezvous point**) and build the tree by sending a packet from each member to the root.

32. Define PIM?

The shared tree approaches like core-based trees are used for multicasting to sparse groups in the Internet as part of popular protocols such as PIM (Protocol Independent Multicast).

33. What is mean by Anycast routing?

Another delivery model, called anycast. In anycast, a packet is delivered to the nearest member of a group. Schemes that find these paths are called anycast routing.

34. Difference between connection-less and connection-oriented service?

Connectionless Service

- In a connectionless service, the packets are sent from one party to another with no need for connection establishment or connection release.
- The packets are not numbered; they may be delayed or lost or may arrive out of sequence.
- There is no acknowledgment either.

Connection-Oriented Service

- In a connection-oriented service, a connection is first established between the sender and the receiver.
- Data are transferred.
- At the end, the connection is released.

35. Define congestion?

Too many packets present in (a part of) the network causes packet delay and loss that degrades performance. This situation is called congestion. The network and transport layers share the responsibility for handling congestion.

36. What is congestion control?

Congestion control refers to the mechanisms and techniques to control the congestion and keep the load below the capacity.

37. Mention the various approaches in congestion control?

- Network provisioning
- Traffic-aware routing
- Admission control
- Traffic throttling
- Load shedding

38. What is network provisioning?

Links and routers that are regularly heavily utilized are upgraded at the earliest opportunity. This is called provisioning and happens on a time scale of months, driven by long-term traffic trends.

39. What is Traffic-aware routing?

Some local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called traffic-aware routing. Splitting traffic across multiple paths is also helpful.

40. What is admission control?

In a virtual-circuit network, new connections can be refused if they would cause the network to become congested. This is called admission control.

41. What is load shedding?

When all fails, the network is forced to discard packets that it cannot deliver. The general name for this is load shedding. A good policy for choosing which packets to discard can help to prevent congestion collapse.

42. Why is admission control required in networks?

(NOV 2015)

One technique that is widely used in virtual-circuit networks to keep congestion at bay is admission control. The idea is simple: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. Thus, attempts to set up a virtual circuit may fail.

43. Why traffic shaping approach is used in congestion.

(MAY 2016)

- First, routers must determine when congestion is approaching, ideally before it has arrived. To do so, each router can continuously monitor the resources it is using.
- The second problem is that routers must deliver timely feedback to the senders that are causing the congestion.

44. Define choke packets?

The router selects a congested packet and sends a choke packet back to the source host, giving it the destination found in the packet.

45. Define RED?

- RED (Random Early Detection) to determine when to start discarding, routers maintain a running average of their queue lengths.
- When the average queue length on some link exceeds a threshold, the link is said to be congested and a small fraction of the packets are dropped at random.
- Picking packets at random makes it more likely that the fastest senders will see a packet drop.

46. Define internetworks?

When two or more networks are connected, they become an internetwork, or internet.

47. List the many ways networks can differ?

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security.	Privacy rules, encryption, etc
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

48. How the networks can be connected?

There are two basic choices for connecting different networks: we can build devices that translate or convert packets from each kind of network into packets for each other network. In either case, the devices are placed at the boundaries between networks.

49. What is multi-protocol router?

A router that can handle multiple network protocols is called a multiprotocol router. It must either translate the protocols, or leave connection for a higher protocol layer.

50. What is mean by Tunneling?

Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4. To pass through this region, the packet must have an IPv4 address.

51. Mention the two-level internetworking routing algorithm?

1. Intra-domain or interior gateway protocol
2. Inter-domain or exterior gateway protocol / BGP (Border Gateway Protocol)

52. Define autonomous system?

Each network is operated independently of all the others, it is often referred to as an AS (Autonomous System). A good mental model for an AS is an ISP network.

53. Define Path MTU?

A source does not usually know the path a packet will take through the network to a destination, so it certainly does not know how small packets must be to get there. This packet size is called the Path MTU (Path Maximum Transmission Unit).

54. Mention the two types of fragmentation?

1. Transparent fragmentation
2. Nontransparent fragmentation

55. Define path MTU discovery?

Each IP packet is sent with its header bits set to indicate that no fragmentation is allowed to be performed. If a router receives a packet that is too large, it generates an error packet, returns it to the source, and drops the packet. The process is called path MTU discovery.

56. What is mean by IPv4?

- The Internet Protocol version 4 (IPv4) is the delivery mechanism used by the TCP/IP protocols.
- IPv4 is an unreliable and connectionless datagram protocol-a best-effort delivery service. The term best-effort means that IPv4 provides no error control or flow control.
- IPv4 must be paired with a reliable protocol such as TCP.
- IPv4 is also a connectionless protocol for a packet-switching network that uses the datagram approach.
- IPv4 is its 32-bit addresses.
- Every host and router on the Internet has an IP address that can be used in the Source address and Destination address fields of IP packets.

57. What is Subnetting?

The solution is to allow the block of addresses to be split into several parts for internal use as multiple networks, while still acting like a single network to the outside world. This is called **subnetting** and the networks (such as Ethernet LANs) that result from dividing up a larger network are called **subnets**.

58. State the difference between classless and classful addressing. (NOV 2016)

- Each router to have the corresponding prefix information. This design works with subnetting and is called CIDR (Classless Inter- Domain Routing).
- IP addresses were divided into the five categories class A, B, C, D and E. This allocation has come to be called classful addressing.
- In classful IP addressing, the network ID portion can take only the predefined number of bits 8, 16, or 24.
- In classless addressing, any number of bits can be assigned to the network ID.

59. What is mean by IPv6?

- IPv6 (Internet Protocol version 6), also known as IPng (Internetworking Protocol, next generation).
- It uses 128-bit addresses.
- In IPv6, the Internet protocol was extensively modified to accommodate the unforeseen growth of the Internet. The format and the length of the IP address were changed along with the packet format.
- The adoption of IPv6 has been slow.

60. Bring out the difference between IPv4 and IPv6 address?**(NOV 2015)**

IPv4	IPv6
IPv4 addresses are 32 bit length	IPv6 addresses are 128 bit length
Binary numbers represented in decimals	Binary numbers represented in hexadecimal
IPSec supports is only optional	Inbuilt IPSec support
Fragmentation is done by sender and forwarding routers	Fragmentation is done only by sender
Checksum field is available	No checksum field
Broadcast messages are available	Broadcast messages are not available

61. What is mean by NAT?

A quick solution to this problem is called network address translation (NAT). NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally. The traffic inside can use the large set; the traffic outside, the small set.

62. List the various IPv6 extension header?

- Hop-by-hop options
- Destination options
- Routing
- Fragmentation
- Authentication
- Encrypted security payload

63. Define ICMP?

- The operation of the Internet is monitored closely by the routers. When something unexpected occurs during packet processing at a router, the event is reported to the sender by the ICMP (Internet Control Message Protocol).
- ICMP is also used to test the Internet.
- Each ICMP message type is carried encapsulated in an IP packet.

64. List the types of ICMP messages?

- Destination unreachable
- Time exceeded
- Parameter problem
- Source quench
- Redirect
- Echo and echo reply
- Timestamp request/reply
- Router advertisement/solicitation

65. Define DHCP?

- It is possible to manually configure each computer, but that is tedious and error-prone. There is a better way, and it is called DHCP (Dynamic Host Configuration Protocol).
- DHCP provides static and dynamic address allocation that can be manual or automatic.
- DHCP, every network must have a DHCP server that is responsible for configuration.
- When a computer is started, it has a built-in Ethernet or other link layer address embedded in the NIC, but no IP address.

66. What is OSPF?

- Open Shortest Path First (OSPF), allow the administrator to assign a cost for passing through a network based on the type of service required. A route through a network can have different costs (metrics).
- OSPF protocol allows each router to have several routing tables based on the required type of service.

67. List the five types of OSPF messages?

- Hello
- Link state update
- Link state ack
- Database description
- Link state request

68. Define BGP?

- Border Gateway Protocol (BGP) is an interdomain routing protocol using path vector routing.
- Within a single AS, OSPF and IS-IS are the protocols that are commonly used. Between ASes, a different protocol, called BGP (Border Gateway Protocol), is used.

69. Find the error, if any in the following IP addresses.**(NOV 2016)****(a) 111.56.045.78 (b) 75.45.301.14****Solution**

- a) There are no leading zeroes in dotted-decimal notation (045).
- b) In dotted-decimal notation, each number is less than or equal to 255; 301 is outside this range.

11 MARKS

1. Explain in detail about the design issues in network layer? (11 MARKS) (MAY 2016)

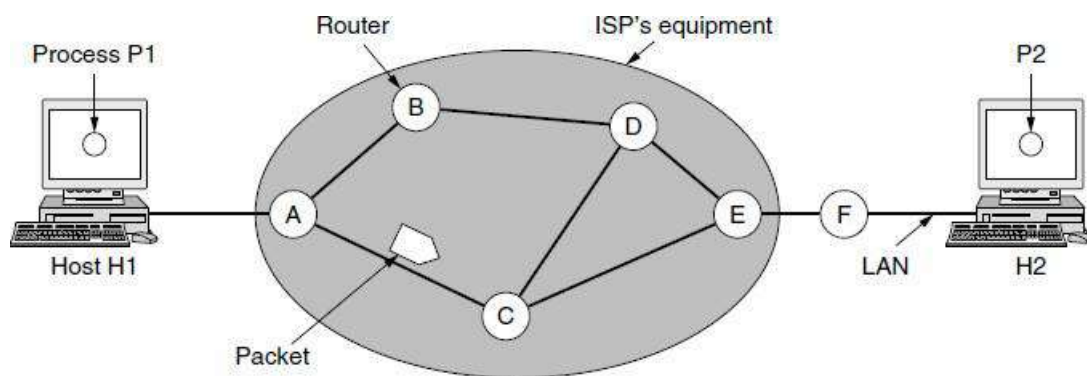
- The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links).
- The network layer is concerned with getting packets from the source all the way to the destination.

The design issues of network layer include the service provided to the transport layer and the internal design of the network.

1. Store-and-Forward Packet Switching
2. Services Provided to the Transport Layer
3. Implementation of Connectionless Service
4. Implementation of Connection-Oriented Service
5. Comparison of Virtual-Circuit and Datagram Subnets

Store-and-Forward Packet Switching

- The major components of the network are the ISP's equipment (routers connected by transmission lines), shown inside the shaded oval, and the customers' equipment, shown outside the oval.
- Host H1 is directly connected to one of the ISP's routers, A, perhaps as a home computer that is plugged into a DSL modem.
- In contrast, H2 is on a LAN, which might be an office Ethernet, with a router, F, owned and operated by the customer. This router has a leased line to the ISP's equipment.
- We have shown F as being outside the oval because it does not belong to the ISP.



The environment of the network layer protocols

- A host with a packet to send transmits it to the nearest router, either on its own LAN or over a point-to-point link to the ISP.
- The packet is stored there until it has fully arrived and the link has finished its processing by verifying the checksum.
- Then it is forwarded to the next router along the path until it reaches the destination host, where it is delivered.

Services Provided to the Transport Layer

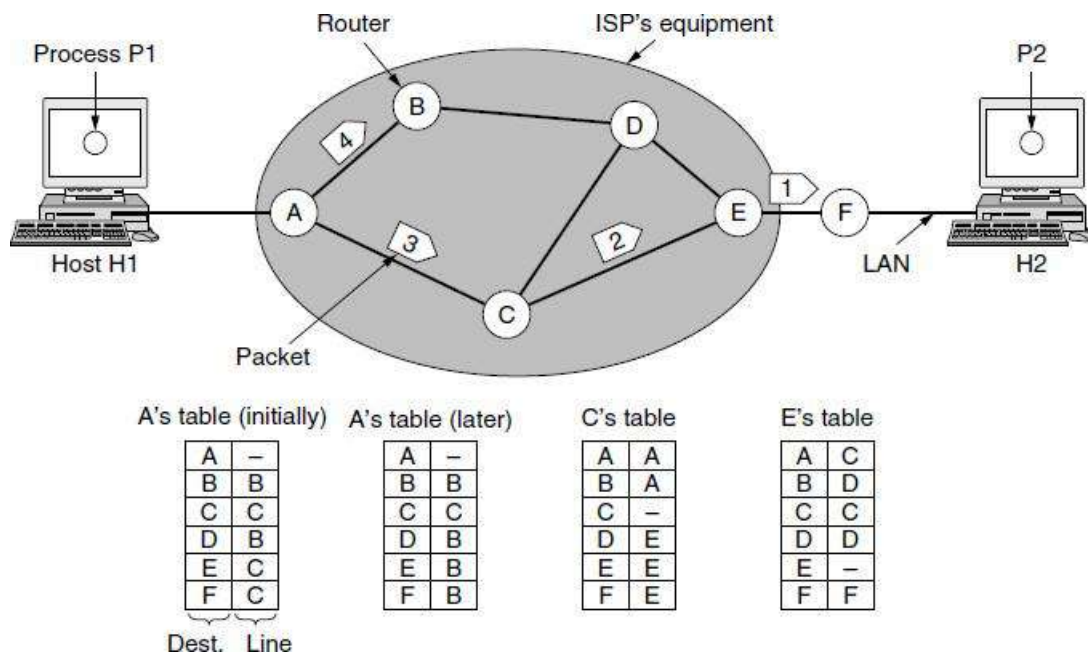
The network layer provides services to the transport layer at the network layer/transport layer interface.

1. The services should be independent of the router technology.
2. The transport layer should be shielded from the number, type, and topology of the routers present.
3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

Implementation of Connectionless Service

- If connectionless service is offered, packets are injected into the network individually and routed independently of each other. No advance setup is needed.
- The packets are frequently called **datagrams** and the network is called a **datagram network**.
- If connection-oriented service is used, a path from the source router all the way to the destination router must be established before any data packets can be sent. This connection is called a **VC (virtual circuit)**, in analogy with the physical circuits set up by the telephone system, and the network is called a **virtual-circuit network**.

Let us now see how a datagram network works



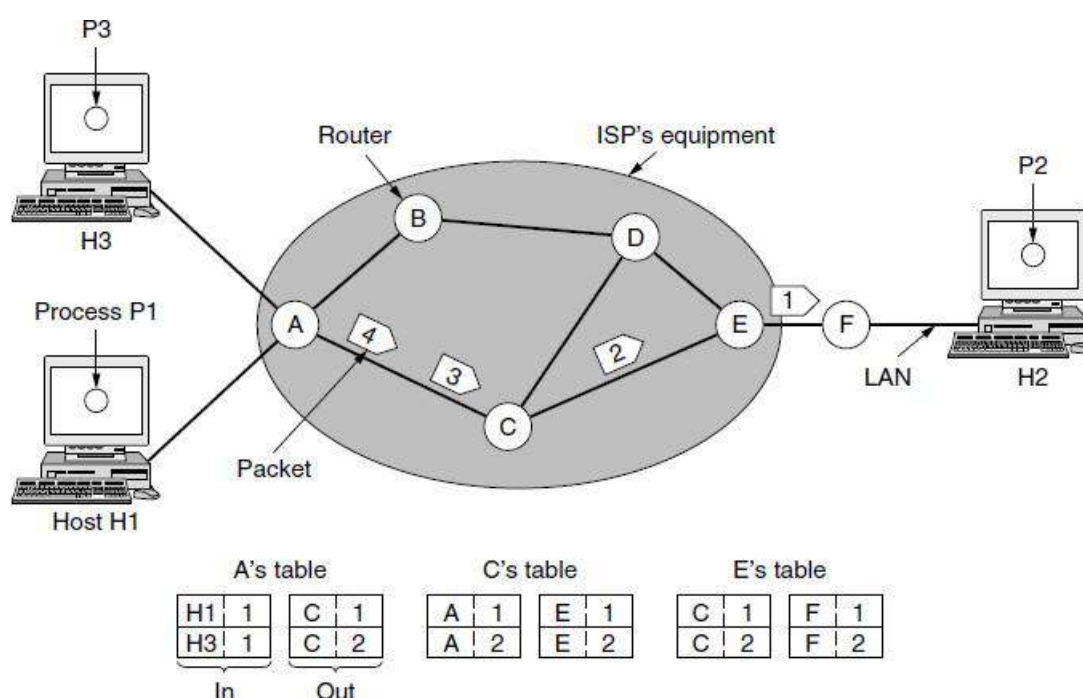
Routing within a datagram network

- Suppose that the process P1 in Figure has a long message for P2. It hands the message to the transport layer, with instructions to deliver it to process P2 on host H2.
- The transport layer code runs on H1, typically within the operating system.
- It prepends a transport header to the front of the message and hands the result to the network layer, probably just another procedure within the operating system.

- IP (Internet Protocol), which is the basis for the entire Internet, is the dominant example of a connectionless network service.
- Each packet carries a destination IP address that routers use to individually forward each packet.
- The addresses are 32 bits in IPv4 packets and 128 bits in IPv6 packets.

Implementation of Connection-Oriented Service

- For connection-oriented service, we need a virtual-circuit network.
- When a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works.
- When the connection is released, the virtual circuit is also terminated.



Routing within a virtual-circuit network

Here, host H1 has established connection 1 with host H2. This connection is remembered as the first entry in each of the routing tables. The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1. Similarly, the first entry at C routes the packet to E, also with connection identifier 1.

An example of a connection-oriented network service is MPLS (Multi-Protocol Label Switching). It is used within ISP networks in the Internet, with IP packets wrapped in an MPLS header having a 20-bit connection identifier or label. MPLS is often hidden from customers, with the ISP establishing long-term connections for large amounts of traffic, but it is increasingly being used to help when quality of service.

Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
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2. Explain in detail about Routing Algorithm? (6 MARKS)

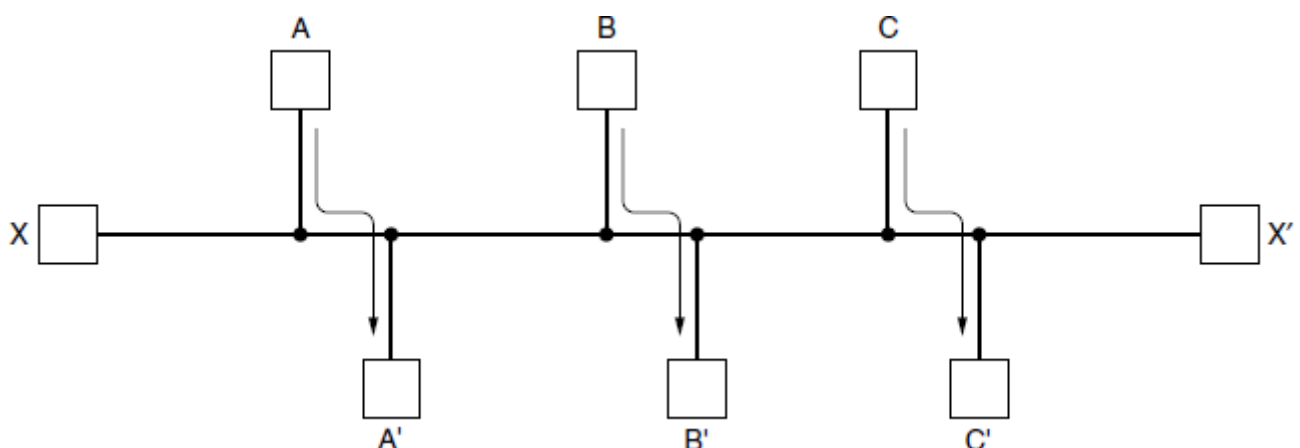
- The **routing algorithm** is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.
- If the network uses datagrams internally, this decision must be made anew for every arriving data packet since the best route may have changed since last time.
- If the network uses virtual circuits internally, routing decisions are made only when a new virtual circuit is being set up.
- Thereafter, data packets just follow the already established route.
- The latter case is sometimes called **session routing** because a route remains in force for an entire session (e.g., while logged in over a VPN).
- The distinction between routing, which is making the decision which routes to use, and forwarding, which is what happens when a packet arrives.
- A router as having two processes inside it. One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. This process is **forwarding**.
- The other process is responsible for filling in and updating the routing tables.

The routes are chosen independently for each packet sent or only when new connections are established, certain properties are desirable in a routing algorithm:

- Correctness,
- Simplicity,
- Robustness,
- Stability,
- Fairness, and
- Efficiency

- **Correctness:** The routing should be done properly and correctly so that the packets may reach their proper destination.
- **Simplicity:** The routing should be done in a simple manner so that the overhead is as low as possible. With increasing complexity of the routing algorithms the overhead also increases.
- **Robustness:** Once a major network becomes operative, it may be expected to run continuously for years without any failures. The algorithms designed for routing should be robust enough to handle hardware and software failures and should be able to cope with changes in the topology and traffic without requiring all jobs in all hosts to be aborted and the network rebooted every time some router goes down.
- **Stability:** The routing algorithms should be stable under all possible circumstances.
- **Fairness:** Every node connected to the network should get a fair chance of transmitting their packets. This is generally done on a first come first serve basis.
- **Efficiency:** The routing algorithms should be optimal in terms of throughput and minimizing mean packet delays.

Suppose that there is enough traffic between A and A', between B and B', and between C and C' to saturate the horizontal links. To maximize the total flow, the X to X' traffic should be shut off altogether. Unfortunately, X and X' may not see it that way. Evidently, some compromise between global efficiency and fairness to individual connections is needed.



Network with a conflict between fairness and efficiency

Routing algorithms can be grouped into two major classes:

1. **Non-adaptive algorithms** and
2. **Adaptive algorithms**

- **Non-adaptive algorithms** do not base their routing decisions on any measurements or estimates of the current topology and traffic. Instead, the choice of the route to use to get from I to J is computed in advance, offline, and downloaded to the routers when the network is booted. This procedure is sometimes called **static routing**.
- Because it does not respond to failures, static routing is mostly useful for situations in which the routing choice is clear.
- **Adaptive algorithms**, in contrast, change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well.
- These **dynamic routing algorithms** differ in where they get their information (e.g., locally, from adjacent routers, or from all routers), when they change the routes (e.g., when the topology changes, or every ΔT seconds as the load changes), and what metric is used for optimization (e.g., distance, number of hops, or estimated transit time).

A variety of routing algorithms are

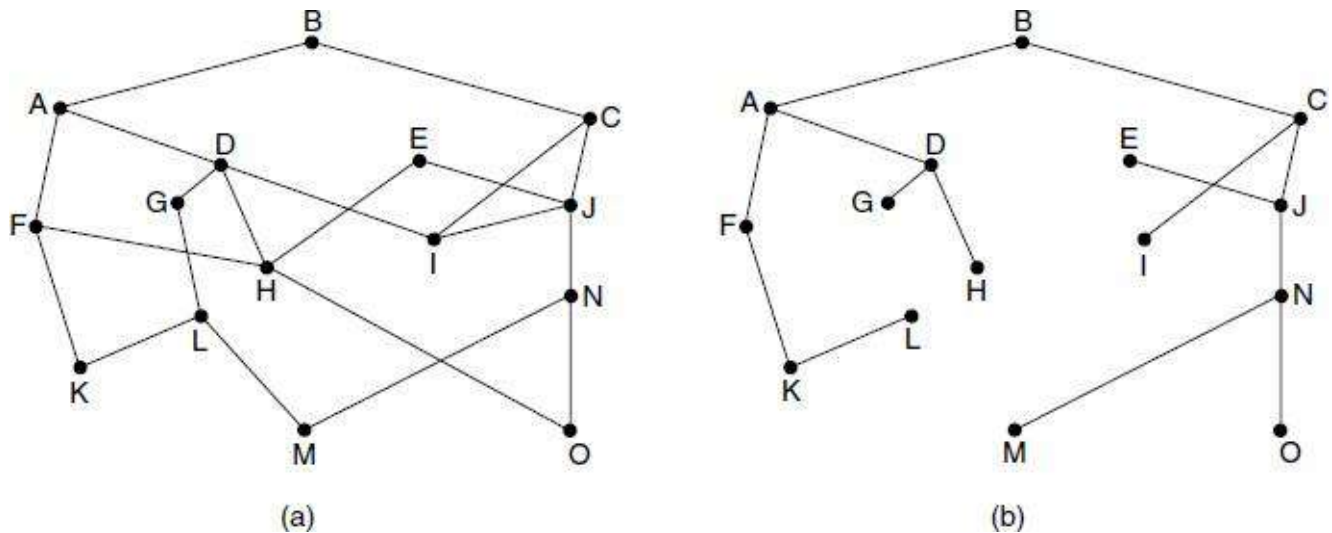
1. **The Optimality Principle**
2. **Shortest Path Routing**
3. **Flooding**
4. **Distance Vector Routing**
5. **Link State Routing**
6. **Hierarchical Routing**
7. **Broadcast Routing**
8. **Multicast Routing**

- The algorithms cover delivery models besides sending a packet from a source to a destination.
- Sometimes the goal is to send the packet to multiple, all, or one of a set of destinations.
- All of the routing algorithms we describe here make decisions based on the topology.

3. Discuss the optimality principle of routing? (6 MARKS) (NOV 2015)

- A general statement about optimal routes without regard to network topology or traffic. This statement is known as the **optimality principle**.
- It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.
- The part of the route from I to J r_1 and the rest of the route r_2 .
- If a route better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K, contradicting our statement that r_1r_2 is optimal.

As a direct consequence of the optimality principle, the set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a **sink tree** and is illustrated in Figure (b), where the distance metric is the number of hops. The goal of all routing algorithms is to discover and use the sink trees for all routers.



(a) A network (b) A sink tree for router B

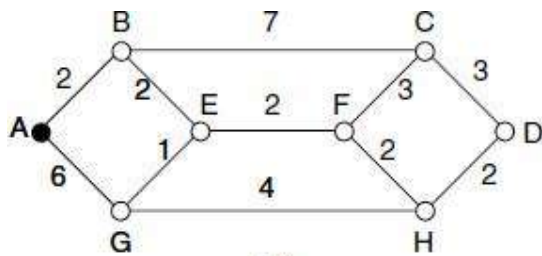
- A sink tree is not necessarily unique; other trees with the same path lengths may exist. If we allow all of the possible paths to be chosen, the tree becomes a more general structure called a DAG (Directed Acyclic Graph). DAGs have no loops.
- We will use sink trees as a convenient shorthand for both cases.
- Both cases also depend on the technical assumption that the paths do not interfere with each other so, for example, a traffic jam on one path will not cause another path to divert.

Since a sink tree is indeed a tree, it does not contain any loops, so each packet will be delivered within a finite and bounded number of hops. Links and routers can go down and come back up during operation, so different routers may have different ideas about the current topology.

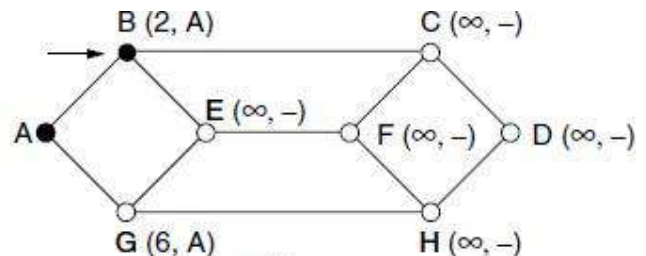
4. Explain in detail about Shortest Path Algorithm? (6 MARKS)

The idea is to build a graph of the network, with each node of the graph representing a router and each edge of the graph representing a communication line, or link. To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.

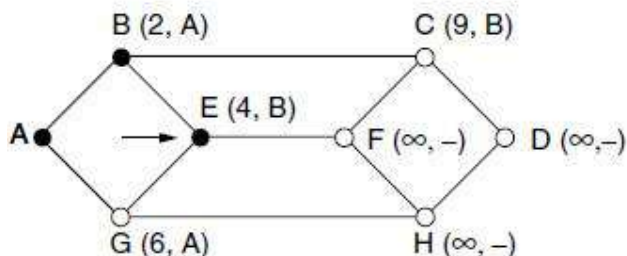
The concept of a **shortest path** deserves as. One way of measuring path length is the number of hops. Using this metric, the paths ABC and ABE in Figure are equally long. Another metric is the geographic distance in kilometers, in which case ABC is clearly much longer than ABE.



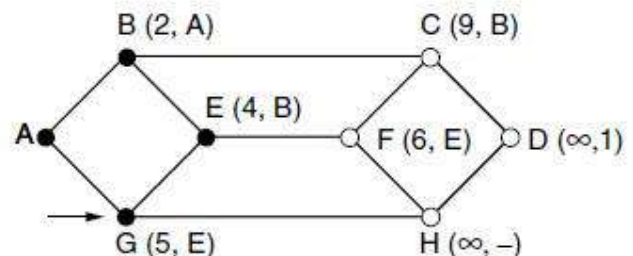
(a)



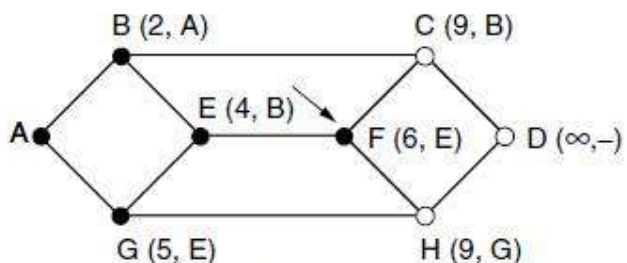
(b)



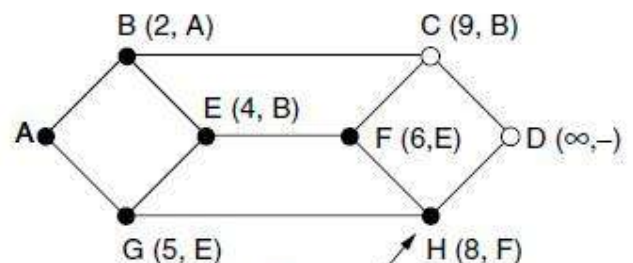
(c)



(d)



(e)



(f)

The first six steps used in computing the shortest path from A to D. The arrows indicate the working node.

- In the general case, the labels on the edges could be computed as a function of the distance, bandwidth, average traffic, communication cost, measured delay, and other factors.
- By changing the weighting function, the algorithm would then compute the “shortest” path measured according to any one of a number of criteria or to a combination of criteria.

Several algorithms for computing the shortest path between two nodes of a graph are known.

- This one is due to Dijkstra and finds the shortest paths between a source and all destinations in the network.
- Each node is labeled (in parentheses) with its distance from the source node along the best known path.
- The distances must be non-negative, as they will be if they are based on real quantities like bandwidth and delay.
- Initially, no paths are known, so all nodes are labeled with infinity.
- As the algorithm proceeds and paths are found, the labels may change, reflecting better paths.
- A label may be either tentative or permanent.
- Initially, all labels are tentative.

- When it is discovered that a label represents the shortest possible path from the source to that node, it is made permanent and never changed thereafter.

To illustrate how the labeling algorithm works, look at the weighted, undirected graph of Figure (a), where the weights represent, for example, distance. We want to find the shortest path from A to D. We start out by marking node A as permanent, indicated by a filled-in circle.

Each of the nodes adjacent to A (the working node), relabeling each one with the distance to A. Whenever a node is relabeled, we also label it with the node from which the probe was made so that we can reconstruct the final path later. If the network had more than one shortest path from A to D and we wanted to find all of them, we would need to remember all of the probe nodes that could reach a node with the same distance.

Each of the nodes adjacent to A, we examine all the tentatively labeled nodes in the whole graph and make the one with the smallest label permanent, as shown in Figure(b). This one becomes the new working node.

We now start at B and examine all nodes adjacent to it. If the sum of the label on B and the distance from B to the node being considered is less than the label on that node, we have a shorter path, so the node is relabeled.

After all the nodes adjacent to the working node have been inspected and the tentative labels changed if possible, the entire graph is searched for the tentatively labeled node with the smallest value. This node is made permanent and becomes the working node for the next round.

To see why the algorithm works, look at Figure(c). At this point we have just made E permanent. Suppose that there were a shorter path than ABE, say AXYZE (for some X and Y). There are two possibilities: either node Z has already been made permanent, or it has not been. If it has, then E has already been probed (on the round following the one when Z was made permanent), so the AXYZE path has not escaped our attention and thus cannot be a shorter path.

Now consider the case where Z is still tentatively labeled. If the label at Z is greater than or equal to that at E, then AXYZE cannot be a shorter path than ABE. If the label is less than that of E, then Z and not E will become permanent first, allowing E to be probed from Z.

Dijkstra's algorithm

```
#define MAX_NODES 1024                                /* maximum number of nodes */
#define INFINITY 1000000000                            /* a number larger than every maximum path */
int n, dist[MAX_NODES][MAX_NODES];                    /* dist[i][j] is the distance from i to j */
void shortest_path(int s, int t, int path[])
{
    struct state                                         /* the path being worked on */
    {
```

```

        int predecessor;                /* previous node */
        int length;                     /* length from source to this node */
        enum { permanent, tentative } label; /* label state */
    } state[MAX_NODES];

    int i, k, min;
    struct state *p;
    for (p = &state[0]; p < &state[n]; p++) /* initialize state */
    {
        p->predecessor = -1;
        p->length = INFINITY;
        p->label = tentative;
    }

    state[t].length = 0; state[t].label = permanent;
    k = t;                /* k is the initial working node */
    do                    /* Is there a better path from k? */
    {
        for (i = 0; i < n; i++) /* this graph has n nodes */
            if (dist[k][i] != 0 && state[i].label == tentative)
            {
                if (state[k].length + dist[k][i] < state[i].length)
                {
                    state[i].predecessor = k;
                    state[i].length = state[k].length + dist[k][i];
                }
            }
    }
    /* Find the tentatively labeled node with the smallest label. */
    k = 0; min = INFINITY;
    for (i = 0; i < n; i++)
        if (state[i].label == tentative && state[i].length < min)
        {
            min = state[i].length;
            k = i;
        }
    state[k].label = permanent;
} while (k != s);

/* Copy the path into the output array. */

```

```

        i = 0; k = s;
    do
    {
        path[i++] = k; k = state[k].predecessor;
    } while (k >= 0);
}

```

Dijkstra's algorithm to compute the shortest path through a graph.

The global variables *n* and *dist* describe the graph and are initialized before shortest path is called. We compute the shortest path starting at the terminal node, *t*, rather than at the source node, *s*.

Since the shortest paths from *t* to *s* in an undirected graph are the same as the shortest paths from *s* to *t*, it does not matter at which end we begin. The reason for searching backward is that each node is labeled with its predecessor rather than its successor. When the final path is copied into the output variable, *path*, the path is thus reversed. The two reversal effects cancel, and the answer is produced in the correct order.

5. Write a short notes on Flooding? (5 MARKS)

- When a routing algorithm is implemented, each router must make decisions based on local knowledge, not the complete picture of the network.
- **Flooding** is a simple computer network routing algorithm in which every incoming packet is sent through every outgoing link except the one it arrived on.
- A simple local technique is **flooding**, in which every incoming packet is sent out on every outgoing line except the one it arrived on.
- Flooding obviously generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the process.
- One such measure is to have a hop counter contained in the header of each packet that is decremented at each hop, with the packet being discarded when the counter reaches zero.
- Ideally, the hop counter should be initialized to the length of the path from source to destination.
- If the sender does not know how long the path is, it can initialize the counter to the worst case, namely, the full diameter of the network.
- Flooding with a hop count can produce an exponential number of duplicate packets as the hop count grows and routers duplicate packets they have seen before.
- A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time.
- One way to achieve this goal is to have the source router put a sequence number in each packet it receives from its hosts.
- Each router then needs a list per source router telling which sequence numbers originating at that source have already been seen.

- If an incoming packet is on the list, it is not flooded.

The important uses of flooding are

- First, it ensures that a packet is delivered to every node in the network. This may be wasteful if there is a single destination that needs the packet, but it is effective for broadcasting information. In wireless networks, all messages transmitted by a station can be received by all other stations within its radio range, which is, in fact, flooding, and some algorithms utilize this property.
- Second, flooding is tremendously robust. Even if large numbers of routers are blown to bits (e.g., in a military network located in a war zone), flooding will find a path if one exists, to get a packet to its destination.
- Flooding also requires little in the way of setup. The routers only need to know their neighbors. This means that flooding can be used as a building block for other routing algorithms that are more efficient but need more in the way of setup.
- Flooding can also be used as a metric against which other routing algorithms can be compared. Flooding always chooses the shortest path because it chooses every possible path in parallel. Consequently, no other algorithm can produce a shorter delay.

6. Explain the working of Distance Vector Routing? (11 MARKS) (NOV 2015, 2016) (MAY 2017)

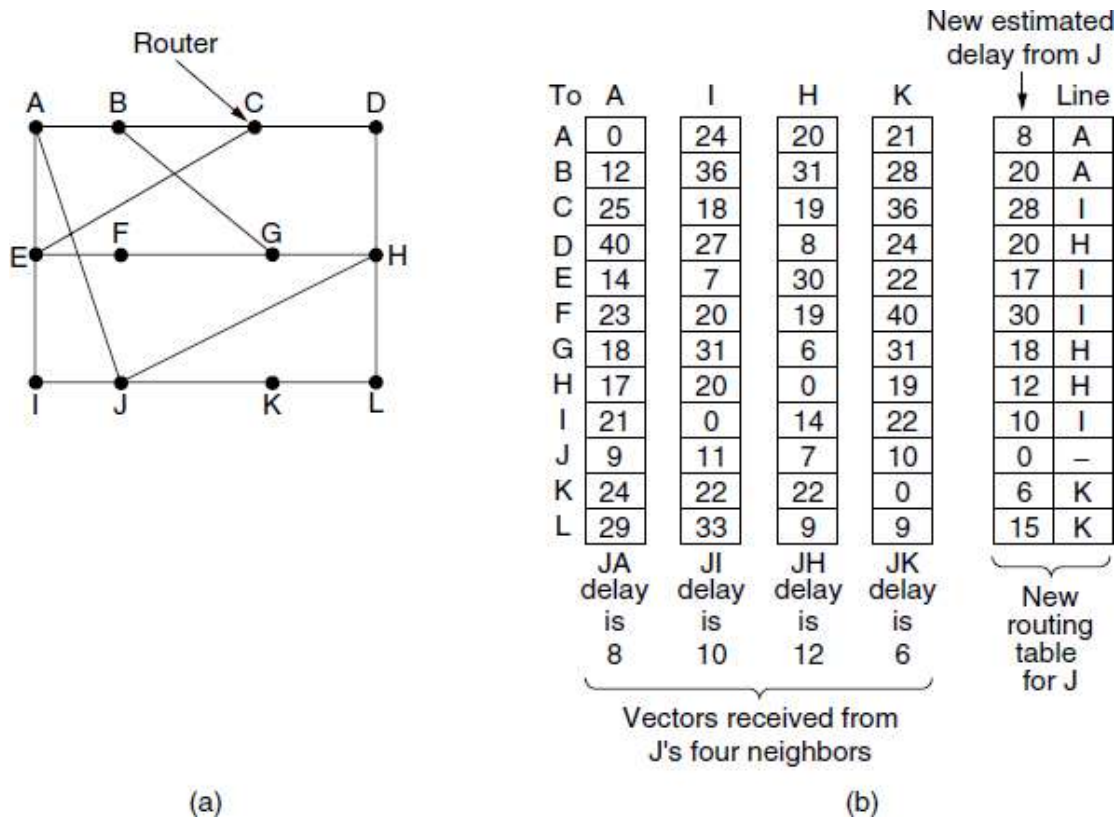
A **distance vector routing** algorithm operates by having each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there. These tables are updated by exchanging information with the neighbors. Eventually, every router knows the best link to reach each destination. The distance vector routing algorithm is sometimes called by other names, most commonly the distributed **Bellman-Ford** routing algorithm.

- In distance vector routing, each router maintains a routing table indexed by, and containing one entry for each router in the network.
 - This entry has two parts: the preferred outgoing line to use for that destination and an estimate of the distance to that destination.
 - The distance might be measured as the number of hops or using another metric, as we discussed for computing shortest paths.
- The router is assumed to know the “distance” to each of its neighbors.
 - If the metric is hops, the distance is just one hop.
 - If the metric is propagation delay, the router can measure it directly with special ECHO packets that the receiver just timestamps and sends back as fast as it can.

As an example, assume that delay is used as a metric and that the router knows the delay to each of its neighbors. Once every T msec, each router sends to each neighbor a list of its estimated delays to each destination. It also receives a similar list from each neighbor.

This updating process is illustrated in Figure.

- Part (a) shows a network. The first four columns of part (b) show the delay vectors received from the neighbors of router J.
- A claims to have a 12-msec delay to B, a 25-msec delay to C, a 40- msec delay to D, etc.
- Suppose that J has measured or estimated its delay to its neighbors, A, I, H, and K, as 8, 10, 12, and 6 msec, respectively.
-



(a) A network. (b) Input from A, I, H, K, and the new routing table for J.

- Consider how J computes its new route to router G. It knows that it can get to A in 8 msec, and furthermore A claims to be able to get to G in 18 msec, so J knows it can count on a delay of 26 msec to G if it forwards packets bound for G to A.
- Similarly, it computes the delay to G via I, H, and K as 41 (31 + 10), 18 (6 + 12), and 37 (31 + 6) msec, respectively.
- The best of these values is 18, so it makes an entry in its routing table that the delay to G is 18 msec and that the route to use is via H.
- The same calculation is performed for all the other destinations, with the new routing table shown in the last column of the figure.

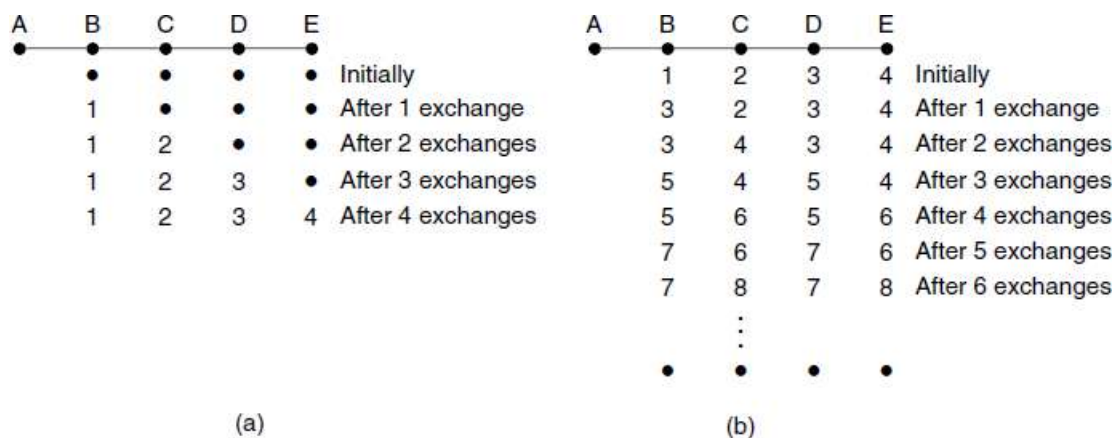
The Count-to-Infinity Problem

- The settling of routes to best paths across the network is called **convergence**.

- Distance vector routing is useful as a simple technique by which routers can collectively compute shortest paths.

Consider a router whose best route to destination X is long. If, on the next exchange, neighbor A suddenly reports a short delay to X, the router just switches over to using the line to A to send traffic to X. In one vector exchange, the good news is processed.

Consider the five-node (linear) network of Figure, where the delay metric is the number of hops. Suppose A is down initially and all the other routers know this. In other words, they have all recorded the delay to A as infinity.



The count-to-infinity problem

When A comes up, the other routers learn about it via the vector exchanges. For simplicity, we will assume that there is a gigantic gong somewhere that is struck periodically to initiate a vector exchange at all routers simultaneously.

At the time of the first exchange, B learns that its left-hand neighbor has zero delay to A. B now makes an entry in its routing table indicating that A is one hop away to the left. All the other routers still think that A is down. At this point, the routing table entries for A are as shown in the second row of Figure (a).

On the next exchange, C learns that B has a path of length 1 to A, so it updates its routing table to indicate a path of length 2, but D and E do not hear the good news until later.

Clearly, spreading at the rate of one hop per exchange. In a network whose longest path is of length N hops, within N exchanges everyone will know about newly revived links and routers.

Now let us consider the situation of Figure (b), in which all the links and routers are initially up. Routers B, C, D, and E have distances to A of 1, 2, 3, and 4 hops, respectively. Suddenly, either A goes down or the link between A and B is cut.

7. Explain briefly about Link State Routing? (11 MARKS) (NOV 2016)

The link state routing is fairly simple and can be stated as five parts. Each router must do the following things to make it work:

- In effect, the complete topology is distributed to every router. Then Dijkstra's algorithm can be run at each router to find the shortest path to every other router.

When a router is booted, its first task is to learn who its neighbors are. It accomplishes this goal by sending a special HELLO packet on each point-to-point line. The router on the other end is expected to send back a reply giving its name.

When two or more routers are connected by a broadcast link (e.g., a switch, ring, or classic Ethernet), the situation is slightly more complicated.

Figure 1.1 shows two network topologies. (a) A hierarchical network with a central LAN connected to three separate clusters. Each cluster has a central node (C, F, H) connected to two peripheral nodes (D, E; G, I; A, B). (b) A more complex network with a central node N connected to A, C, and F. A, C, and F are further connected to other nodes, forming a mesh-like structure. A router is indicated at node H.

(a) Nine routers and a broadcast LAN. (b) A graph model of (a).

The broadcast LAN provides connectivity between each pair of attached routers. However, modeling the LAN as many point-to-point links increases the size of the topology and leads to wasteful messages. A better way to model the LAN is to consider it as a node itself, as shown in Figure(b).

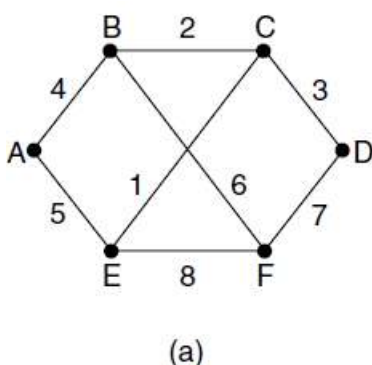
Here, we have introduced a new, artificial node, N, to which A, C, and F are connected. One designated router on the LAN is selected to play the role of N in the routing protocol. The fact that it is possible to go from A to C on the LAN is represented by the path ANC here.

Setting Link Costs

- The link state routing algorithm requires each link to have a distance or cost metric for finding shortest paths.
- The cost to reach neighbors can be set automatically, or configured by the network operator.
- A common choice is to make the cost inversely proportional to the bandwidth of the link.
- For example, 1-Gbps Ethernet may have a cost of 1 and 100-Mbps Ethernet a cost of 10. This makes higher-capacity paths better choices.
- If the network is geographically spread out, the delay of the links may be factored into the cost so that paths over shorter links are better choices.
- The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately.
- By measuring the round-trip time and dividing it by two, the sending router can get a reasonable estimate of the delay.

Building Link State Packets

- Once the information needed for the exchange has been collected, the next step is for each router to build a packet containing all the data.
- The packet starts with the identity of the sender, followed by a sequence number and age and a list of neighbors.
- The cost to each neighbor is also given.
- An example network is presented in Figure (a) with costs shown as labels on the lines.
- The corresponding link state packets for all six routers are shown in Figure (b).



Link		State		Packets	
A	B	C	D	E	F
Seq.	Seq.	Seq.	Seq.	Seq.	Seq.
Age	Age	Age	Age	Age	Age
B 4	A 4	B 2	C 3	A 5	B 6
E 5	C 2	D 3	F 7	C 1	D 7
	F 6	E 1		F 8	E 8

(b)

(a) A network. (b) The link state packets for this network.

Distributing the Link State Packets

- The trickiest part of the algorithm is distributing the link state packets.
- All of the routers must get all of the link state packets quickly and reliably.
- If different routers are using different versions of the topology, the routes they compute can have inconsistencies such as loops, unreachable machines, and other problems.

The basic distribution algorithm as

- The fundamental idea is to use flooding to distribute the link state packets to all routers.
- To keep the flood in check, each packet contains a sequence number that is incremented for each new packet sent.
- Routers keep track of all the (source router, sequence) pairs.
- When a new link state packet comes in, it is checked against the list of packets already seen.
 - If it is new, it is forwarded on all lines except the one it arrived on.
 - If it is a duplicate, it is discarded.
 - If a packet with a sequence number lower than the highest one seen so far ever arrives, it is rejected as being obsolete as the router has more recent data.

This algorithm has a few problems, but they are manageable.

1. First, if the sequence numbers wrap around, confusion will reign. The solution here is to use a 32-bit sequence number. With one link state packet per second, it would take 137 years to wrap around, so this possibility can be ignored.
2. Second, if a router ever crashes, it will lose track of its sequence number. If it starts again at 0, the next packet it sends will be rejected as a duplicate.
3. Third, if a sequence number is ever corrupted and 65,540 is received instead of 4 (a 1-bit error), packets 5 through 65,540 will be rejected as obsolete, since the current sequence number will be thought to be 65,540.

The solution to all these problems is to include the age of each packet after the sequence number and decrement it once per second. When the age hits zero, the information from that router is discarded. Normally, a new packet comes in, say, every 10 sec, so router information only times out when a router is down.

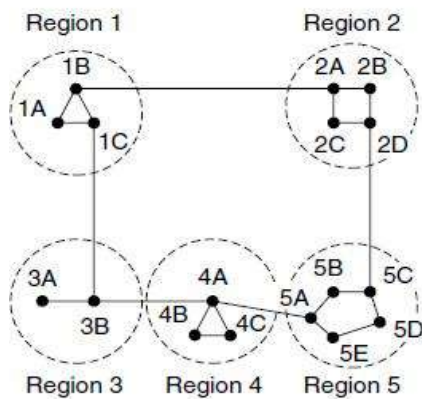
The Age field is also decremented by each router during the initial flooding process, to make sure no packet can get lost and live for an indefinite period of time (a packet whose age is zero is discarded).

Computing the New Routes

- Once a router has accumulated a full set of link state packets, it can construct the entire network graph because every link is represented.
- Every link is, in fact, represented twice, once for each direction.
- The different directions may even have different costs. The shortest-path computations may then find different paths from router A to B than from router B to A.
- Dijkstra's algorithm can be run locally to construct the shortest paths to all possible destinations.
- The results of this algorithm tell the router which link to use to reach each destination. This information is installed in the routing tables, and normal operation is resumed.
- Many ISPs use the **IS-IS (Intermediate System-Intermediate System)** link state protocol.
- It was designed for an early network called DECnet, later adopted by ISO for use with the OSI protocols and then modified to handle other protocols as well, most notably, IP.
- **OSPF (Open Shortest Path First)** is the other main link state protocol. It was designed by IETF several years after IS-IS and adopted many of the innovations designed for IS-IS.

8. Write a short notes on Hierarchical Routing? (5 MARKS)

- A hierarchical routing technique in which the entire address space is divided into levels based on specific criteria.
- When hierarchical routing is used, the routers are divided into what we will call **regions**.
- Each router knows all the details about how to route packets to destinations within its own region but knows nothing about the internal structure of other regions.
- When different networks are interconnected, it is natural to regard each one as a separate region to free the routers in one network from having to know the topological structure of the other ones.
- An example of routing in a two-level hierarchy with five regions. The full routing table for router 1A has 17 entries, as shown in Figure (b).
- When routing is done hierarchically, as in Figure(c), there are entries for all the local routers, as before, but all other regions are condensed into a single router, so all traffic for region 2 goes via the 1B-2A line, but the rest of the remote traffic goes via the 1C-3B line.
- Hierarchical routing has reduced the table from 17 to 7 entries. As the ratio of the number of regions to the number of routers per region grows, the savings in table space increase.
- For example, the best route from 1A to 5C is via region 2, but with hierarchical routing all traffic to region 5 goes via region 3, because that is better for most destinations in region 5.



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

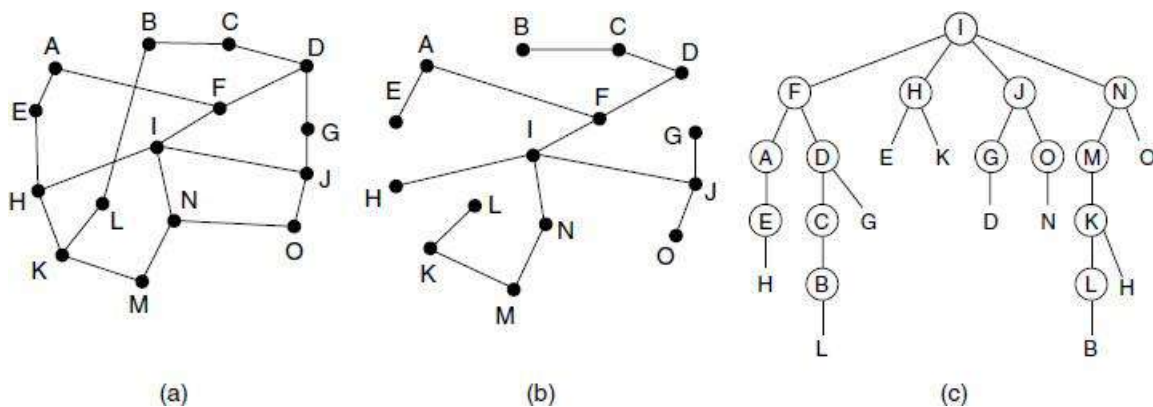
(c)

Hierarchical routing

9. Write a short notes on Broadcast Routing? (11 MARKS)

- In broadcast communication, the relationship between the source and the destination is one-to-all. There is only one source, but all the other hosts are the destinations.
 - In some applications, hosts need to send messages to many or all other hosts.
 - For example, a service distributing weather reports, stock market updates, or live radio programs might work best by sending to all machines and read the data.
 - Sending a packet to all destinations simultaneously is called **broadcasting**.
 - One broadcasting method that requires no special features from the network is for the source to simply send a distinct packet to each destination.
- An improvement is **multi-destination routing**, in which each packet contains either a list of destinations or a bit map indicating the desired destinations.
 - When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed.
 - The router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line.
 - In effect, the destination set is partitioned among the output lines.
 - After a sufficient number of hops, each packet will carry only one destination like a normal packet.
 - Multi-destination routing is like using separately addressed packets, except that when several packets must follow the same route.
 - The network bandwidth is therefore used more efficiently.
 - However the source to know all the destinations, plus it is as much work for a router to determine where to send one multi-destination packet as it is for multiple distinct packets.

- A better broadcast routing technique: flooding.
 - When implemented with a sequence number per source, flooding uses links efficiently with a decision rule at routers that is relatively simple.
 - Although flooding is ill-suited for ordinary point-to-point communication.
- The idea for **reverse path forwarding** is elegant.
 - When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets toward the source of the broadcast.
 - The broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router.
 - The router forwards copies of it onto all links except the one it arrived on.
 - If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.



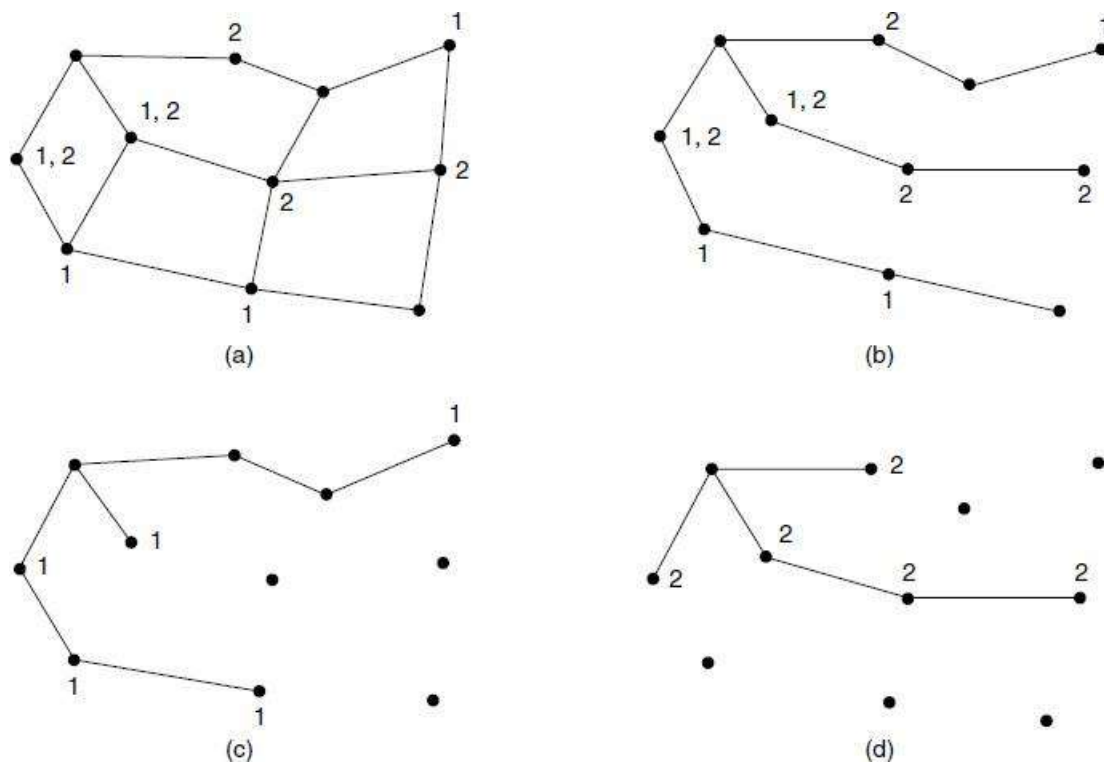
Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

- An example of reverse path forwarding is shown in Figure Part (a) shows a network, part (b) shows a sink tree for router I of that network, and part (c) shows how the reverse path algorithm works.
- On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree.
- Each of these packets arrives on the preferred path to I and is so indicated by a circle around the letter.
- On the second hop, eight packets are generated, two by each of the routers that received a packet on the first hop.
- As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line.
- Of the six packets generated on the third hop, only three arrive on the preferred path (at C, E, and K); the others are duplicates.
- After five hops and 24 packets, the broadcasting terminates, compared with four hops and 14 packets had the sink tree been followed exactly.

- A **spanning tree** is a subset of the network that includes all the routers but contains no loops. Sink trees are spanning trees.
- If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.
- This method makes excellent use of bandwidth, generating the absolute minimum number of packets.

10. Explain briefly about Multicast Routing? (11 MARKS)

- Sending a message to such a group is called **multicasting**, and the routing algorithm used is called **multicast routing**.
- All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group.
- Multicast routing schemes build on the broadcast routing schemes, sending packets along spanning trees to deliver the packets to the members of the group while making efficient use of bandwidth.
- However, the best spanning tree to use depends on whether the group is dense, with receivers scattered over most of the network, with much of the network not belonging to the group.



(a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

- As an example, consider the two groups, 1 and 2, in the network shown in Figure (a). Some routers are attached to hosts that belong to one or both of these groups, as indicated in the figure.
- A spanning tree for the leftmost router is shown in Figure (b). This tree can be used for broadcast but is overkill for multicast.

- In Figure(c), all the links that do not lead to hosts that are members of group 1 have been removed. The result is the multicast spanning tree for the leftmost router to send to group 1. Packets are forwarded only along this spanning tree, which is more efficient than the broadcast tree because there are 7 links instead of 10.
- Figure (d) shows the multicast spanning tree after pruning for group 2. It is efficient too, with only five links this time. It also shows that different multicast groups have different spanning trees.

Various ways of pruning the spanning tree are possible

- **Multicast Open Shortest Path First (MOSPF)** protocol is an extension of the OSPF protocol that uses multicast link state routing to create source-based trees.
- The **Distance Vector Multicast Routing Protocol (DVMRP)** is an implementation of multicast distance vector routing.
- An alternative design uses **core-based trees** to compute a single spanning tree for the group.
- All of the routers agree on a root (called the **core** or **rendezvous point**) and build the tree by sending a packet from each member to the root. The tree is the union of the paths traced by these packets.
- The shared tree approaches like core-based trees are used for multicasting to sparse groups in the Internet as part of popular protocols such as **Protocol Independent Multicast (PIM)**.

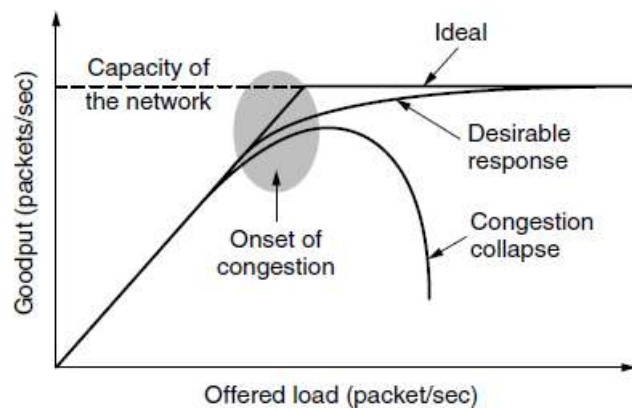
11. Explain in detail about Congestion Control Algorithms? (6 MARKS)

- Too many packets present in the network causes packet delay and loss that degrades performance. This situation is called **congestion**.
- The network and transport layers share the responsibility for handling congestion.
- Since congestion occurs within the network, it is the network layer that directly experiences it and must ultimately determine what to do with the excess packets.
- However, the most effective way to control congestion is to reduce the load that the transport layer is placing on the network.
- This requires the network and transport layers to work together.

When the number of packets hosts send into the network is well within its carrying capacity, the number delivered is proportional to the number sent. If twice as many are sent, twice as many are delivered.

However, as the offered load approaches the carrying capacity, bursts of traffic occasionally fill up the buffers inside routers and some packets are lost. These lost packets consume some of the capacity, so the number of delivered packets falls below the ideal curve. The network is now congested.

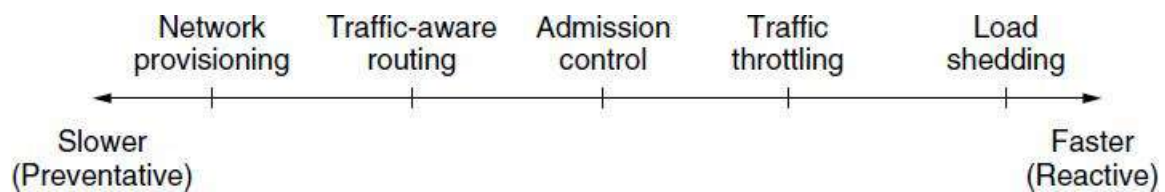
A **congestion collapse**, in which performance plummets as the offered load increases beyond the capacity. This can happen because packets can be sufficiently delayed inside the network that they are no longer useful when they leave the network.



With too much traffic, performance drops sharply

Approaches to Congestion Control

- The presence of congestion means that the load is (temporarily) greater than the resources (in a part of the network) can handle.
- Two solutions come to mind:
 - increase the resources or
 - decrease the load
- As shown in Figure, these solutions are usually applied on different time scales to either prevent congestion or react to it once it has occurred.



Timescales of approaches to congestion control

- The most basic way to avoid congestion is to build a network that is well matched to the traffic that it carries.
- If there is a low-bandwidth link on the path along which most traffic is directed, congestion is likely.
- Sometimes resources can be added dynamically when there is serious congestion, for example, turning on spare routers or enabling lines that are normally used only as backups or purchasing bandwidth on the open market.
- More often, links and routers that are regularly heavily utilized are upgraded at the earliest opportunity. This is called **provisioning** and happens on a time scale of months, driven by long-term traffic trends.
- To make the most of the existing network capacity, routes can be tailored to traffic patterns that change during the day as network user's wake and sleep in different time zones.

- Some local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called **traffic-aware routing**. Splitting traffic across multiple paths is also helpful.

However, sometimes it is not possible to increase capacity. The only way then to beat back the congestion is to decrease the load. In a virtual-circuit network, new connections can be refused if they would cause the network to become congested. This is called **admission control**.

Finally, when all else fails, the network is forced to discard packets that it cannot deliver. The general name for this is **load shedding**.

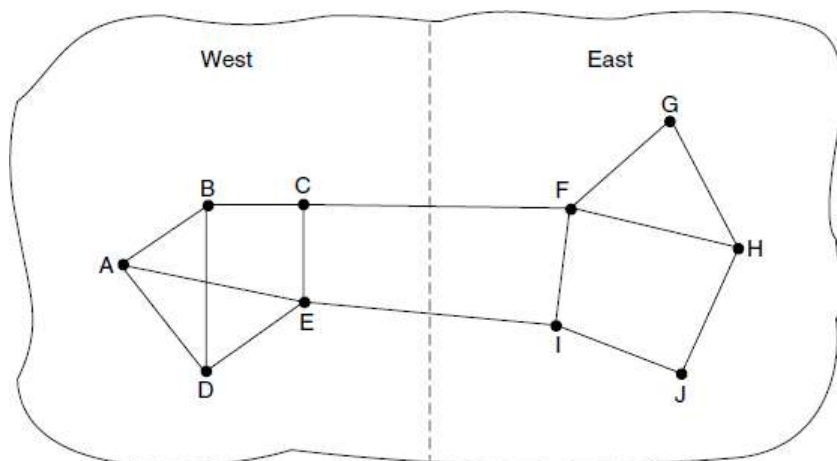
12. Explain the various approaches in congestion control algorithm? (11 MARKS) (NOV 2015)

The various approaches in congestion control algorithms are

1. Traffic-aware routing
2. Admission control
3. Traffic Throttling
4. Load shedding

1. Traffic-aware routing

- Traffic-aware routing was used in the early Internet according to this model.
- Consider the network of Figure, which is divided into two parts, East and West, connected by two links, CF and EI.
- Suppose that most of the traffic between East and West is using link CF, and, as a result, this link is heavily loaded with long delays.
- Including queueing delay in the weight used for the shortest path calculation will make EI more attractive.
- After the new routing tables have been installed, most of the East-West traffic will now go over EI, loading this link.
- Consequently, in the next update, CF will appear to be the shortest path.
- As a result, the routing tables may oscillate wildly, leading to erratic routing and many potential problems.



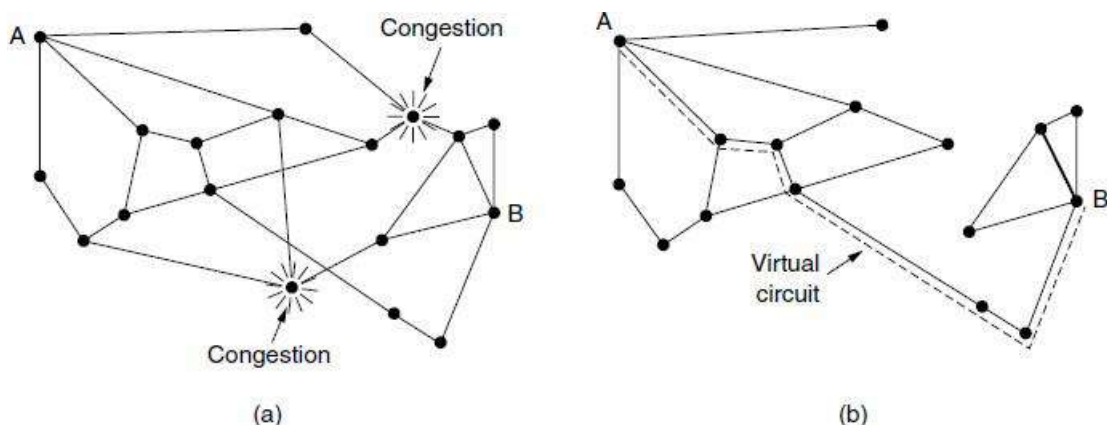
A network in which the East and West parts are connected by two links

In the Internet routing protocols do not generally adjust their routes depending on the load. Instead, adjustments are made outside the routing protocol by slowly changing its inputs. This is called **traffic engineering**.

2. Admission Control

- One technique that is widely used in virtual-circuit networks to keep congestion at bay is **admission control**.
- The idea is simple: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested.
- Thus, attempts to set up a virtual circuit may fail.
- In the telephone system, when a switch gets overloaded it practices admission control by not giving dial tones.
- The trick with this approach is working out when a new virtual circuit will lead to congestion.
- The task is straightforward in the telephone network because of the fixed bandwidth of calls (64 kbps for uncompressed audio).
- However, virtual circuits in computer networks come in all shapes and sizes.
- Traffic is often described in terms of its rate and shape.
- For example, traffic that varies while browsing the Web is more difficult to handle than a streaming movie with the same long-term throughput because the bursts of Web traffic are more likely to congest routers in the network.
- A commonly used descriptor that captures this effect is the **leaky bucket** or **token bucket**.
- A leaky bucket has two parameters that bound the average rate and the instantaneous burst size of traffic.
- Since leaky buckets are widely used for quality of service.

Admission control can also be combined with traffic-aware routing by considering routes around traffic hotspots as part of the setup procedure. For example, consider the network illustrated in Figure (a), in which two routers are congested, as indicated.



(a) A congested network. (b) The portion of the network that is not congested. A virtual circuit from A to B is also shown.

Suppose that a host attached to router A wants to set up a connection to a host attached to router B. Normally, this connection would pass through one of the congested routers. To avoid this situation, we can redraw the network as shown in Figure (b), omitting the congested routers and all of their lines. The dashed line shows a possible route for the virtual circuit that avoids the congested routers.

3. Traffic Throttling

Throttling traffic that can be used in both datagram networks and virtual-circuit networks. Each approach must solve two problems.

1. First, routers must determine when congestion is approaching, ideally before it has arrived. Each router can continuously monitor the resources it is using. Three possibilities are the utilization of the output links, the buffering of queued packets inside the router, and the number of packets that are lost due to insufficient buffering.
2. The second one is the most useful. Averages of utilization do not directly account for the burstiness of most traffic—a utilization of 50% may be low for smooth traffic and too high for highly variable traffic. Counts of packet losses come too late. Congestion has already set in by the time that packets are lost.

The queueing delay inside routers directly captures any congestion experienced by packets. It should be low most of time, but will jump when there is a burst of traffic that generates a backlog.

To maintain a good estimate of the queueing delay, d , a sample of the instantaneous queue length, s , can be made periodically and d updated according to

$$d_{\text{new}} = \alpha d_{\text{old}} + (1 - \alpha)s$$

where the constant α determines how fast the router forgets recent history. This is called a **EWMA (Exponentially Weighted Moving Average)**. It smoothes out fluctuations and is equivalent to a low-pass filter. Whenever d moves above the threshold, the router notes the onset of congestion.

The second problem is that routers must deliver timely feedback to the senders that are causing the congestion. Congestion is experienced in the network, but relieving congestion requires action on behalf of the senders that are using the network. To deliver feedback, the router must identify the appropriate senders.

Choke Packets

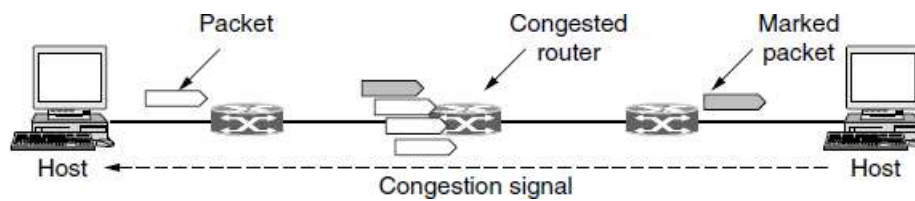
- The most direct way to notify a sender of congestion is to tell it directly.
- In this approach, the router selects a congested packet and sends a **choke packet** back to the source host, giving it the destination found in the packet.
- The original packet may be tagged (a header bit is turned on) so that it will not generate any more choke packets farther along the path and then forwarded in the usual way.
- To avoid increasing load on the network during a time of congestion, the router may only send choke packets at a low rate.
- An example of a choke packet used in the early Internet is the SOURCE-QUENCH message.

Explicit Congestion Notification

ECN (Explicit Congestion Notification) and is used in the Internet. It is a refinement of early congestion signaling protocols. Two bits in the IP packet header are used to record whether the packet has experienced congestion.

Packets are unmarked when they are sent, as illustrated in Figure. If any of the routers they pass through is congested, that router will then mark the packet as having experienced congestion as it is forwarded.

The destination will then echo any marks back to the sender as an explicit congestion signal in its next reply packet. This is shown with a dashed line in the figure to indicate that it happens above the IP level (e.g., in TCP). The sender must then throttle its transmissions, as in the case of choke packets.

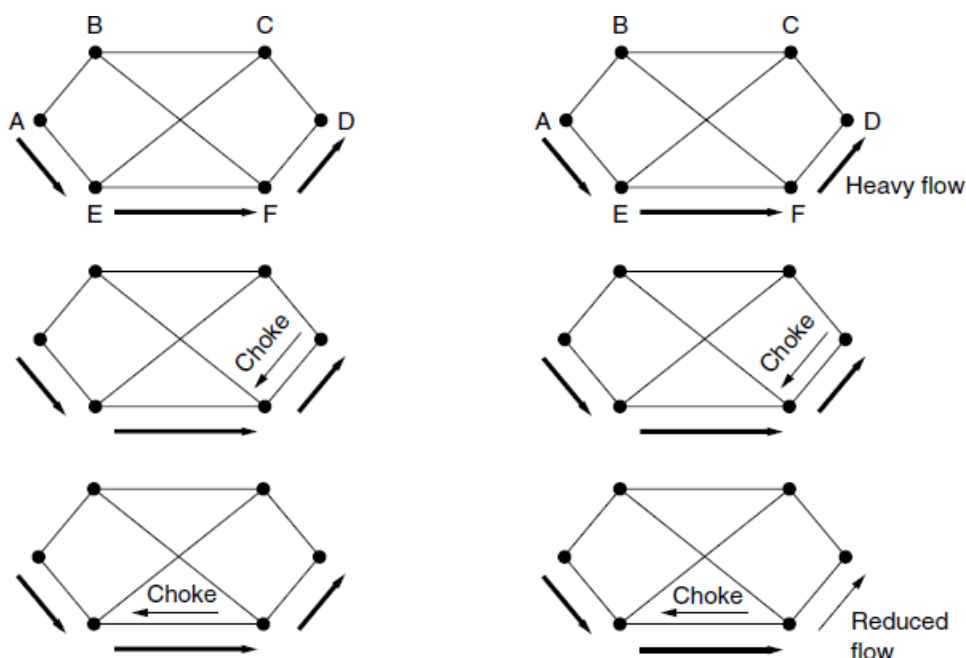


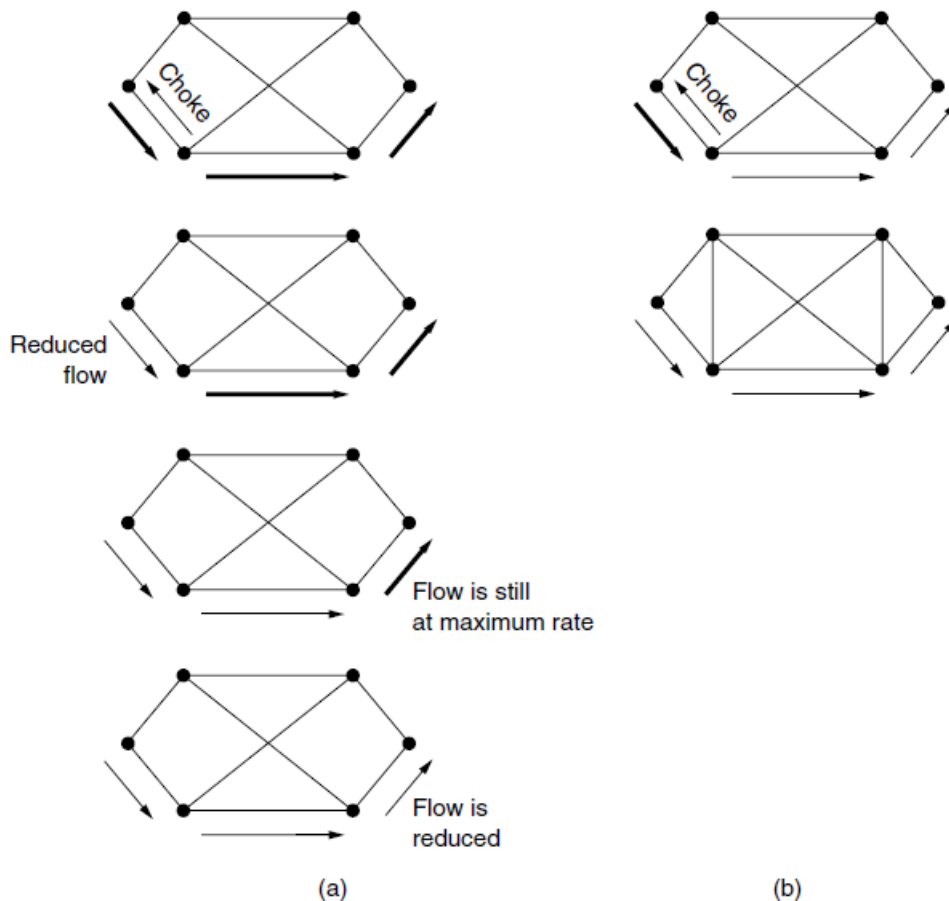
Explicit congestion notification

Hop-by-Hop Backpressure

At high speeds or over long distances, many new packets may be transmitted after congestion has been signaled because of the delay before the signal takes effect.

Consider, for example, a host in San Francisco (router A) that is sending traffic to a host in New York (router D) at the OC-3 speed of 155 Mbps. If the New York host begins to run out of buffers, it will take about 40 msec for a choke packet to get back to San Francisco to tell it to slow down. An ECN indication will take even longer because it is delivered via the destination.





(a) A choke packet that affects only the source. (b) A choke packet that affects each hop it passes through.

The net effect of this hop-by-hop scheme is to provide quick relief at the point of congestion, at the price of using up more buffers upstream. In this way, congestion can be nipped in the bud without losing any packets.

5. Load Shedding

- Load shedding is that when routers are being inundated by packets that they cannot handle, they just throw them away.
- For a router drowning in packets is which packets to drop.
- The preferred choice may depend on the type of applications that use the network.

More intelligent load shedding requires cooperation from the senders.

- An example is packets that carry routing information. These packets are more important than regular data packets because they establish routes; if they are lost, the network may lose connectivity.
- Another example is that algorithms for compressing video, like MPEG, periodically transmit an entire frame and then send subsequent frames as differences from the last full frame.
- In this case, dropping a packet that is part of a difference is preferable to dropping one that is part of a full frame because future packets depend on the full frame.

To implement an intelligent discard policy, applications must mark their packets to indicate to the network how important they are. Then, when packets have to be discarded, routers can first drop packets from the least important class, then the next most important class, and so on.

Random Early Detection

- The popular algorithm is called **RED (Random Early Detection)**.
- To determine when to start discarding, routers maintain a running average of their queue lengths.
- When the average queue length on some link exceeds a threshold, the link is said to be congested and a small fraction of the packets are dropped at random.
- Picking packets at random makes it more likely that the fastest senders will see a packet drop; this is the best option since the router cannot tell which source is causing the most trouble in a datagram network. The affected sender will notice the loss when there is no acknowledgement, and then the transport protocol will slow down.
- The lost packet is thus delivering the same message as a choke packet, but implicitly, without the router sending any explicit signal.
- RED routers improve performance compared to routers that drop packets only when their buffers are full, though they may require tuning to work well.
- For example, the ideal number of packets to drop depends on how many senders need to be notified of congestion.

13. Explain briefly about Internetworking and also explain the networks differ? (11 MARKS) (MAY 2016)

- When two or more networks are connected to form an **internetwork**, or more simply an **internet**.
- To join networks together if everyone used a single networking technology, and it is often the case that there is a dominant kind of network, such as Ethernet.

The Internet is the prime example of this interconnection.

- The purpose of joining all these networks is to allow users on any of them to communicate with users on all the other ones.
- When you pay an ISP for Internet service, you may be charged depending on the bandwidth of your line, but what you are really paying for is the ability to exchange packets with any other host that is also connected to the Internet.

The approach used so successfully by IP (Internet Protocol), the network layer protocol of the Internet, including techniques for

- tunneling through networks,
- routing in internetworks, and
- packet fragmentation

How Networks Differ

- Networks can differ in many ways. Some of the differences, such as different modulation techniques or frame formats, are internal to the physical and data link layers.
- The differences that can be exposed to the network layer.
- It is papering over these differences that makes internetworking more difficult than operating within a single network.

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

Some of the many ways networks can differ

When packets sent by a source on one network must transit one or more foreign networks before reaching the destination network, many problems can occur at the interfaces between networks. To start with, the source needs to be able to address the destination.

What do we do if the source is on an Ethernet network and the destination is on a WiMAX network?

Assuming we can even specify a WiMAX destination from an Ethernet network, packets would cross from a connectionless network to a connection-oriented one. This may require that a new connection be set up on short notice, which injects a delay, and much overhead if the connection is not used for many more packets.

For example, a gateway joining two networks might generate separate packets for each destination in lieu of better network support for multicasting. A large packet might be broken up, sent in pieces, and then joined back together. Receivers might buffer packets and deliver them in order.

Networks also can differ in large respects that are more difficult to reconcile. The clearest example is **quality of service**.

If one network has strong QoS and the other offers best effort service, it will be impossible to make bandwidth and delay guarantees for real-time traffic end to end. In fact, they can likely only be made while the best-effort network is operated at a low utilization, or hardly used, which is unlikely to be the goal of most ISPs.

Security mechanisms are problematic, but at least encryption for confidentiality and data integrity can be layered on top of networks.

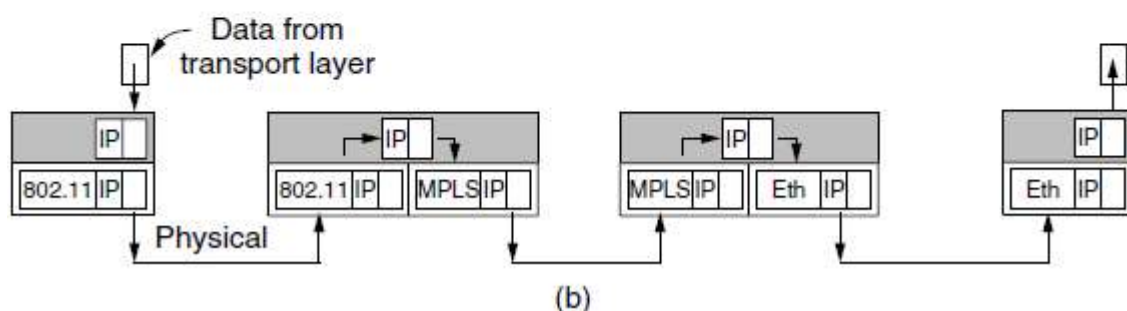
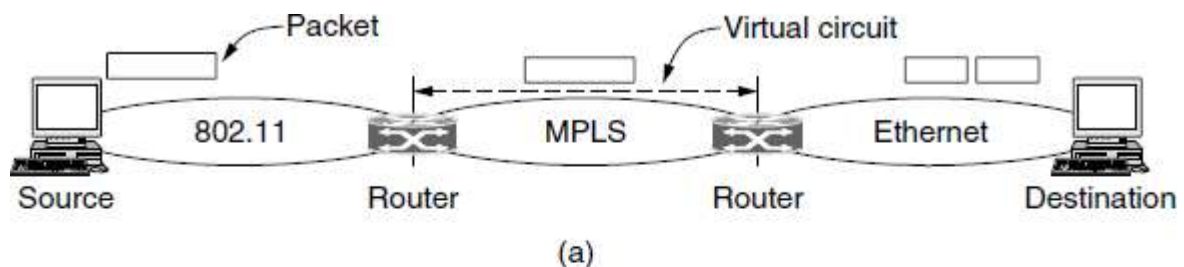
How Networks Can Be Connected

There are two basic choices for connecting different networks:

1. We can build devices that translate or convert packets from each kind of network into packets for each other network, to solve the problem by adding a layer of indirection and building a common layer on top of the different networks.
2. In either case, the devices are placed at the boundaries between networks.

Let us first explore at a high level how interconnection with a common network layer can be used to interconnect dissimilar networks.

- An internet comprised of 802.11, MPLS, and Ethernet networks is shown in Figure (a).
- Suppose that the source machine on the 802.11 network wants to send a packet to the destination machine on the Ethernet network.
- Since these technologies are different, and they are further separated by another kind of network (MPLS), some added processing is needed at the boundaries between the networks.



(a) A packet crossing different networks. (b) Network and link layer protocol processing.

The protocol processing for this journey is shown in Figure (b). The source accepts data from the transport layer and generates a packet with the common network layer header, which is IP in this example. The network header contains the ultimate destination address, which is used to determine that the packet should be sent via the first router.

So the packet is encapsulated in an 802.11 frame whose destination is the first router and transmitted. At the router, the packet is removed from the frame's data field and the 802.11 frame header is discarded. The router now examines the IP address in the packet and looks up this address in its routing table.

Based on this address, it decides to send the packet to the second router next. For this part of the path, an MPLS virtual circuit must be established to the second router and the packet must be encapsulated with MPLS headers that travel this circuit.

At the far end, the MPLS header is discarded and the network address is again consulted to find the next network layer hop. It is the destination itself.

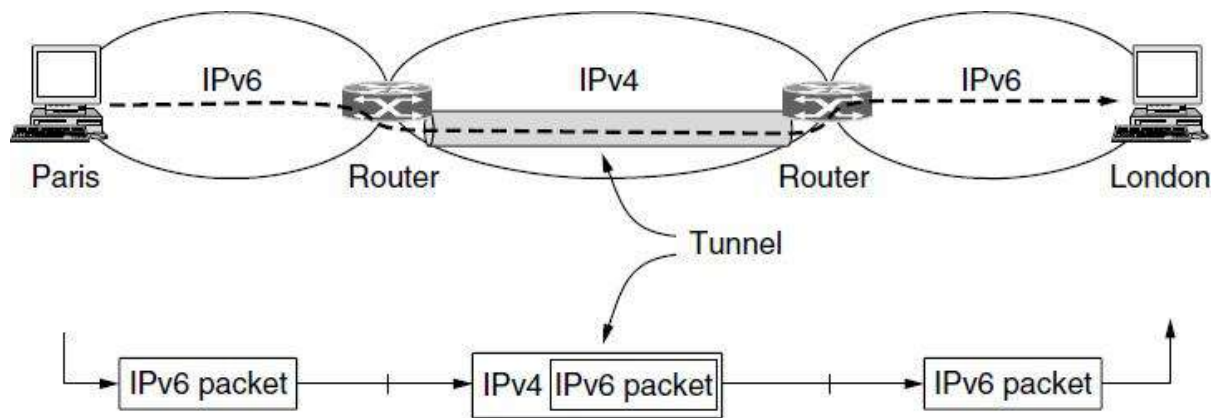
Since the packet is too long to be sent over Ethernet, it is split into two portions. Each of these portions is put into the data field of an Ethernet frame and sent to the Ethernet address of the destination. At the destination, the Ethernet header is stripped from each of the frames, and the contents are reassembled. The packet has finally reached its destination.

Internetworking has been very successful at building large networks, but it only works when there is a common network layer. Examples besides IP, which is now the near-universal network protocol, were IPX, SNA, and AppleTalk.

A router that can handle multiple network protocols is called a multiprotocol router. It must either translate the protocols, or leave connection for a higher protocol layer. Connection at a higher layer using TCP, requires that all the networks implement TCP. Then, it limits usage across the networks to applications that use TCP.

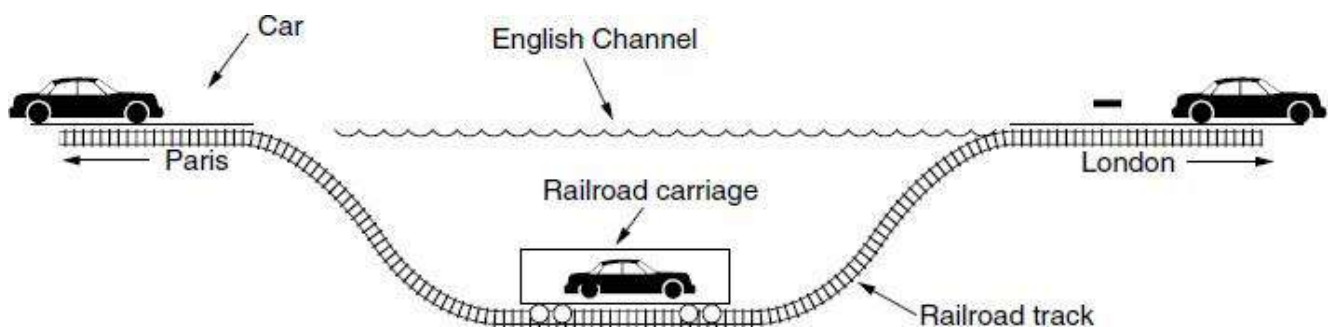
14. Write a short note on Tunneling? (5 MARKS)

- Handling the general case of making two different networks interwork is exceedingly difficult.
- However, there is a common special case that is manageable even for different network protocols.
- The source and destination hosts are on the same type of network, but there is a different network in between.
- As an example, think of an international bank with an IPv6 network in Paris, an IPv6 network in London and connectivity between the offices via the IPv4 Internet.
- The solution to this problem is a technique called tunneling. To send an IP packet to a host in the London office, a host in the Paris office constructs the packet containing an IPv6 address in London, and sends it to the multiprotocol router that connects the Paris IPv6 network to the IPv4 Internet.
- When this router gets the IPv6 packet, it encapsulates the packet with an IPv4 header addressed to the IPv4 side of the multiprotocol router that connects to the London IPv6 network. That is, the router puts a (IPv6) packet inside a (IPv4) packet. When this wrapped packet arrives, the London router removes the original IPv6 packet and sends it onward to the destination host.



Tunneling a packet from Paris to London

Consider a person driving her car from Paris to London. Within France, the car moves under its own power, but when it hits the English Channel, it is loaded onto a high-speed train and transported to England through the Chunnel. Effectively, the car is being carried as freight, as depicted in Figure. At the far end, the car is let loose on the English roads and once again continues to move under its own power. Tunneling of packets through a foreign network works the same way.



Tunneling a car from France to England

Tunneling is widely used to connect isolated hosts and networks using other networks. The network that results is called an overlay since it has effectively been overlaid on the base network. Deployment of a network protocol with a new feature is a common reason, as our “IPv6 over IPv4” example shows.

The disadvantage of tunneling is that none of the hosts on the network that is tunneled over can be reached because the packets cannot escape in the middle of the tunnel.

However, this limitation of tunnels is turned into an advantage with **VPNs (Virtual Private Networks)**.

15. Write a short notes on Internetwork Routing? (5 MARKS)

- Routing through an internet poses the same basic problem as routing within a single network.
- To start, the networks may internally use different routing algorithms.

- For example, one network may use link state routing and another distance vector routing.
- Since link state algorithms need to know the topology but distance vector algorithms.

All of these considerations lead to a two-level routing algorithm.

1. Within each network, an **intradomain or interior gateway protocol** is used for routing.
2. Across the networks that make up the internet, an **interdomain or exterior gateway protocol** is used. The networks may all use different intradomain protocols, but they must use the same interdomain protocol.
3. In the Internet, the interdomain routing protocol is called **BGP (Border Gateway Protocol)**.

Since each network is operated independently of all the others, it is often referred to as an **AS (Autonomous System)**.

- In the Internet, a large determining factor is the business arrangements between ISPs. Each ISP may charge or receive money from the other ISPs for carrying traffic.
- All of these nontechnical factors are wrapped up in the concept of a **routing policy** that governs the way autonomous networks select the routes that they use.

16. Explain in detail about Packet Fragmentation? (11 MARKS)

(MAY 2016)

Each network or link imposes some maximum size on its packets.

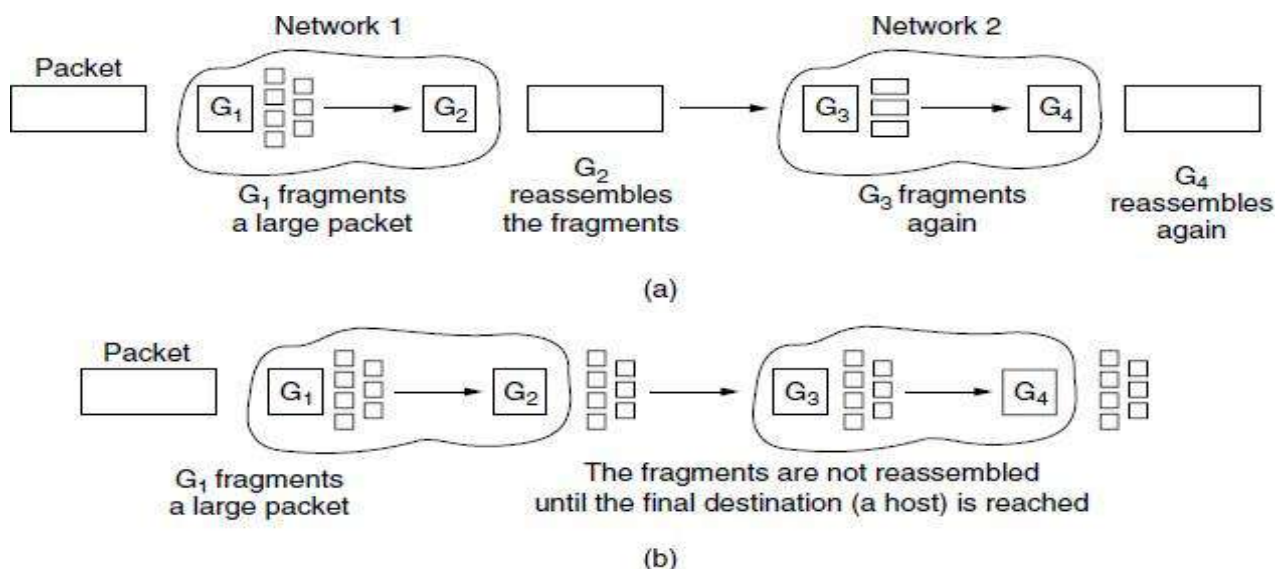
These limits have various causes, among them

1. Hardware (e.g., the size of an Ethernet frame)
2. Operating system (e.g., all buffers are 512 bytes)
3. Protocols (e.g., the number of bits in the packet length field)
4. Compliance with some inter-national standard
5. Desire to reduce error-induced retransmissions to some level
6. Desire to prevent one packet from occupying the channel too long

A source does not usually know the path a packet will take through the network to a destination, so it certainly does not know how small packets must be to get there. This packet size is called the **Path MTU (Path Maximum Transmission Unit)**.

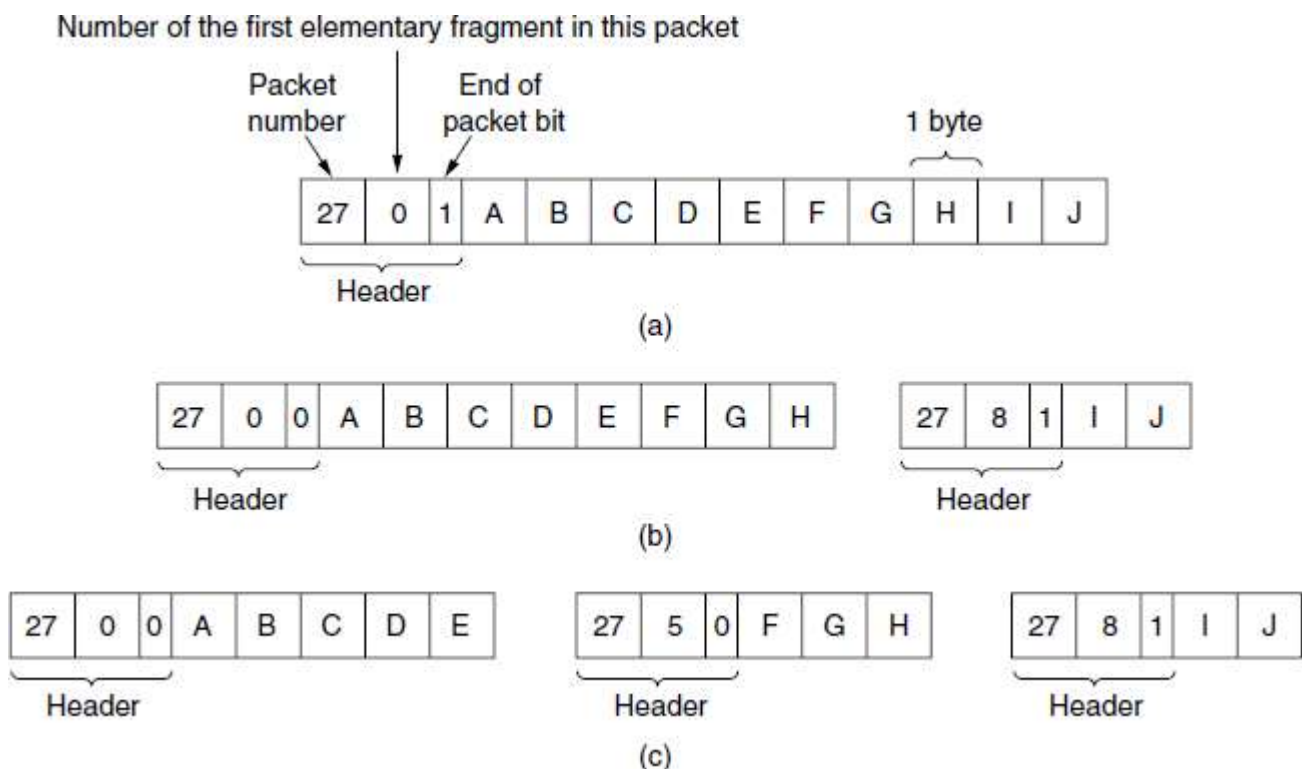
The alternative solution to the problem is to allow routers to break up packets into fragments, sending each fragment as a separate network layer packet. Converting a large object into small fragments is considerably easier than the reverse process. Packet-switching networks, too, have trouble putting the fragments back together again.

Two opposing strategies exist for recombining the fragments back into the original packet. The first strategy is to make fragmentation caused by a “small packet” network transparent to any subsequent networks through which the packet must pass on its way to the ultimate destination. This option is shown in Figure (a). In this approach, when an oversized packet arrives at G1, the router breaks it up into fragments. Each fragment is addressed to the same exit router, G2, where the pieces are recombined. In this way, passage through the small-packet network is made transparent. Subsequent networks are not even aware that fragmentation has occurred.



(a) Transparent fragmentation. (b) Nontransparent fragmentation.

The other fragmentation strategy is to refrain from recombining fragments at any intermediate routers. Once a packet has been fragmented, each fragment is treated as though it were an original packet. The routers pass the fragments, as shown in Figure (b), and reassembly is performed only at the destination host.



Fragmentation when the elementary data size is 1 byte. (a) Original packet, containing 10 data bytes. (b) Fragments after passing through a network with maximum packet size of 8 payload bytes plus header. (c) Fragments after passing through a size 5 gateway.

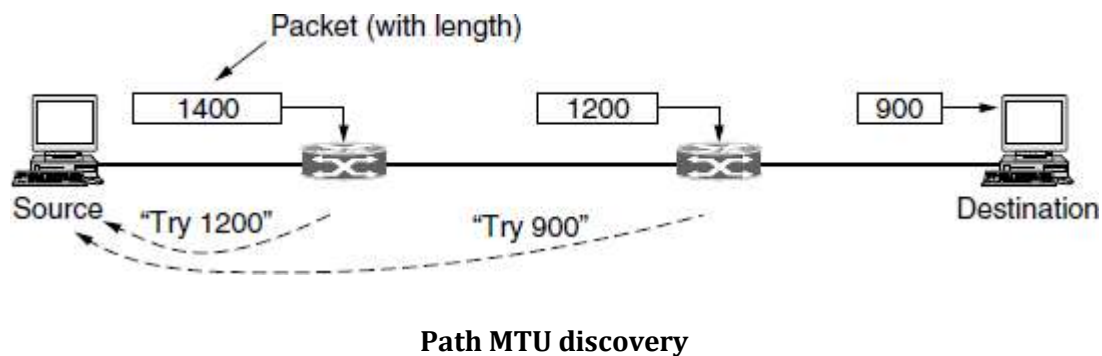
The design used by IP is to give every fragment a packet number (carried on all packets), an absolute byte offset within the packet, and a flag indicating whether it is the end of the packet. An example is shown in Figure.

Fragments can be placed in a buffer at the destination in the right place for reassembly, even if they arrive out of order. Fragments can also be fragmented if they pass over a network with a yet smaller MTU. This is shown in Figure (c). Retransmissions of the packet (if all fragments were not received) can be fragmented into different pieces.

Finally, fragments can be of arbitrary size, down to a single byte plus the packet header. In all cases, the destination simply uses the packet number and fragment offset to place the data in the right position, and the end-of-packet flag to determine when it has the complete packet.

The original solution of getting rid of fragmentation in the network is the strategy used in the modern Internet. The process is called **path MTU discovery**.

Each IP packet is sent with its header bits set to indicate that no fragmentation is allowed to be performed. If a router receives a packet that is too large, it generates an error packet, returns it to the source, and drops the packet. This is shown in Figure.

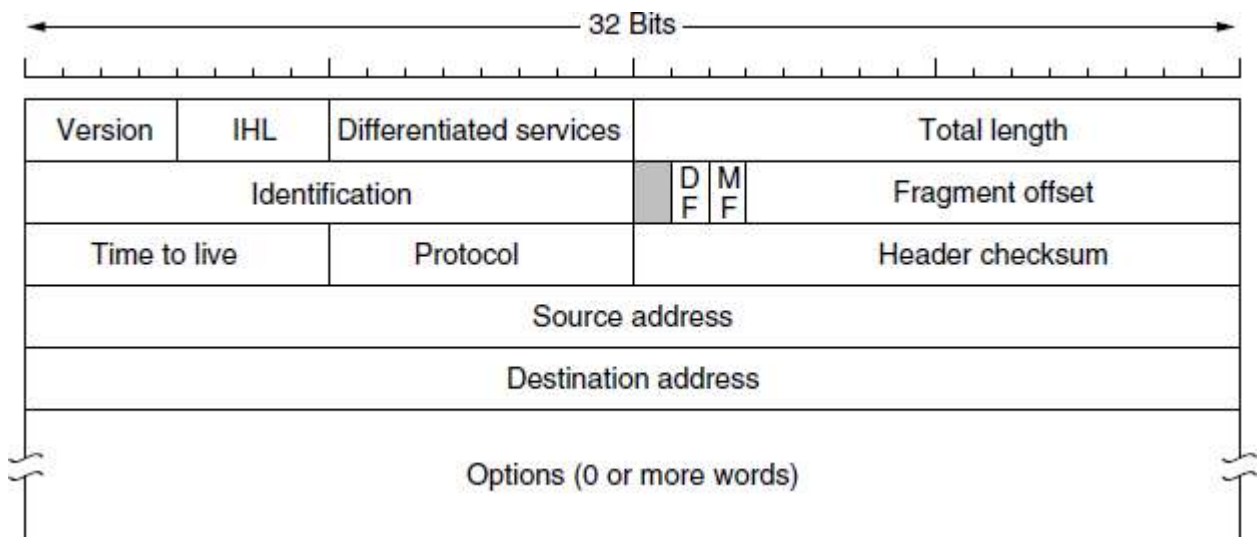


When the source receives the error packet, it uses the information inside to refragment the packet into pieces that are small enough for the router to handle. If a router further down the path has an even smaller MTU, the process is repeated.

7. Explain briefly about IPv4 in detail? (6 MARKS) (NOV 2016)

- The Internet Protocol (IP) is the method or protocol by which data is sent from one computer to another on the Internet.
- Each computer (known as a host) on the Internet has at least one IP address that uniquely identifies it from all other computers on the Internet.
- The Internet Protocol version 4 (IPv4) is the delivery mechanism used by the TCP/IP protocols.

An IPv4 datagram consists of a header part and a body or payload part. The header has a 20-byte fixed part and a variable-length optional part. The header format is shown in Figure.



The IPv4 (Internet Protocol) header

- The bits are transmitted from left to right and top to bottom, with the high-order bit of the **Version field**.
- The **Version field** keeps track of which version of the protocol the datagram belongs to.
- The version at the start of each datagram, it becomes possible to have a transition between versions over a long period of time.
- Since the header length is not constant, a field in the header, **IHL**, is provided to how long the header is, in 32-bit words.
- The minimum value is 5, which applies when no options are present.
- The maximum value of this 4-bit field is 15, which limits the header to 60 bytes, and thus the **Options field** to 40 bytes.
- The Differentiated services field is called the **Type of service field**.
- The **Type of service** field provided 3 bits to signal priority and 3 bits to signal whether a host cared more about delay, throughput, or reliability.
- The **Total length** includes everything in the datagram—both header and data. The maximum length is 65,535 bytes.
- The **Identification field** is needed to allow the destination host to determine which packet a newly arrived fragment belongs to. All the fragments of a packet contain the same Identification value.
- Then come two 1-bit fields related to fragmentation. **DF** stands for **Don't Fragment**. It is an order to the routers not to fragment the packet.
- **MF** stands for **More Fragments**. All fragments except the last one have this bit set.
- The **Fragment offset** tells where in the current packet this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit. Since 13 bits are provided, there is a maximum of 8192 fragments per datagram, supporting a maximum packet length up to the limit of the **Total length field**.

- The **TtL (Time to live) field** is a counter used to limit packet lifetimes. It was originally supposed to count time in seconds, allowing a maximum lifetime of 255 sec. It must be decremented on each hop and is supposed to be decremented multiple times when a packet is queued for a long time in a router.
- The **Protocol field** tells it which transport process to give the packet to. TCP is one possibility, but so are UDP and some others. The numbering of protocols is global across the entire Internet.
- Since the header carries vital information such as addresses, it rates its own checksum for protection, the **Header checksum**. The algorithm is to add up all the 16-bit half-words of the header as they arrive, using one's complement arithmetic, and then take the one's complement of the result. The **Header checksum** is assumed to be zero upon arrival. Such a checksum is useful for detecting errors while the packet travels through the network.
- The **Source address** and **Destination address** indicate the IP address of the source and destination network interfaces.
- The **Options field** was designed to provide an escape to allow subsequent versions of the protocol to include information.
- The options are of variable length.
- Each begins with a 1-byte code identifying the option.
- Some options are followed by a 1-byte option length field, and then one or more data bytes.
- The Options field is padded out to a multiple of 4 bytes.

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Some of the IP options

18. Explain briefly about IP Addresses? (11 MARKS)

- A defining feature of IPv4 is its 32-bit addresses.
- Every host and router on the Internet has an IP address that can be used in the Source address and Destination address fields of IP packets.
- It really refers to a network interface, so if a host is on two networks, it must have two IP addresses. However, most hosts are on one network and thus have one IP address.
- In contrast, routers have multiple interfaces and thus multiple IP addresses.

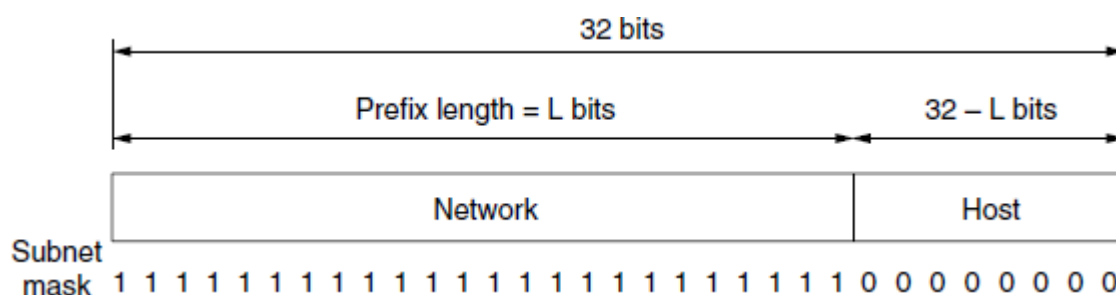
Prefixes

- IP addresses are hierarchical, unlike Ethernet addresses.
- Each 32-bit address is comprised of a variable-length network portion in the top bits and a host portion in the bottom bits.

- The network portion has the same value for all hosts on a single network, such as an Ethernet LAN.
- This means that a network corresponds to a contiguous block of IP address space. This block is called a **prefix**.

IP addresses are written in **dotted decimal notation**. In this format, each of the 4 bytes is written in decimal, from 0 to 255. For example, the 32-bit hexadecimal address 80D00297 is written as 128.208.2.151. Prefixes are written by giving the lowest IP address in the block and the size of the block. The size is determined by the number of bits in the network portion; the remaining bits in the host portion can vary.

Since the prefix length cannot be inferred from the IP address alone, routing protocols must carry the prefixes to routers. Sometimes prefixes are simply described by their length, as in a “/16” which is pronounced “slash 16.” The length of the prefix corresponds to a binary mask of 1s in the network portion. When written out this way, it is called a **subnet mask**. It can be ANDed with the IP address to extract only the network portion. For our example, the subnet mask is 255.255.255.0.



An IP prefix and a subnet mask.

Hierarchical addresses have significant advantages.

- The key advantage of prefixes is that routers can forward packets based on only the network portion of the address, as long as each of the networks has a unique address block.
- The host portion does not matter to the routers because all hosts on the same network will be sent in the same direction.
- When the packets reach the network for which they are destined that they are forwarded to the correct host.

While using a hierarchy lets Internet routing scale, it has two disadvantages

- First, the IP address of a host depends on where it is located in the network. An Ethernet address can be used anywhere in the world, but every IP address belongs to a specific network, and routers will only be able to deliver packets destined to that address to the network.
- The second disadvantage is that the hierarchy is wasteful of addresses unless it is carefully managed. If addresses are assigned to networks in (too) large blocks, there will be (many) addresses that are allocated.

Subnets

Network numbers are managed by a nonprofit corporation called **ICANN (Internet Corporation for Assigned Names and Numbers)**, to avoid conflicts. In turn, ICANN has delegated parts of the address space to various regional authorities, which dole out IP addresses to ISPs and other companies. This is the process by which a company is allocated a block of IP addresses.

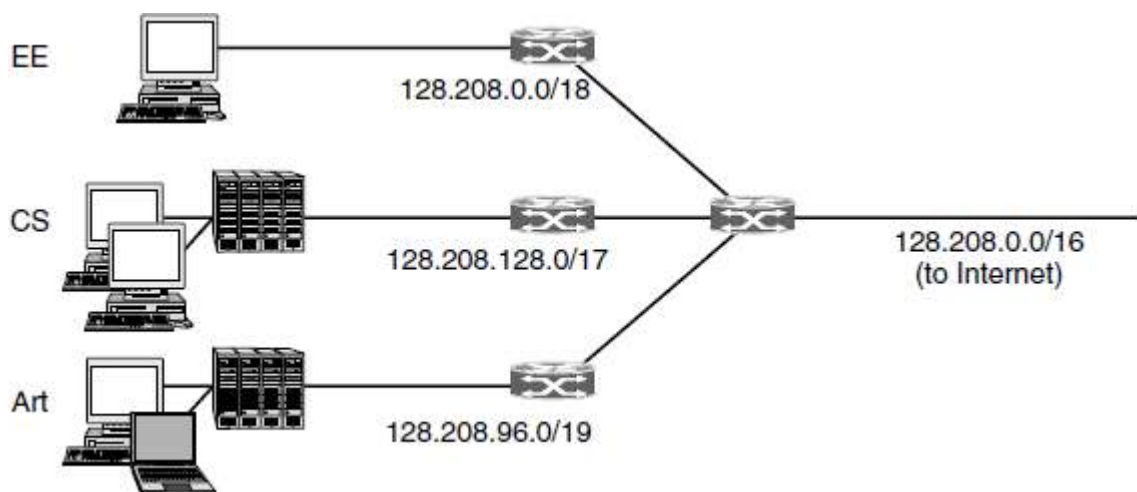
The solution is to allow the block of addresses to be split into several parts for internal use as multiple networks, while still acting like a single network to the outside world. This is called **subnetting** and the networks (such as Ethernet LANs) that result from dividing up a larger network are called **subnets**.

Figure shows how subnets can help with our example. The single /16 has been split into pieces. This split does not need to be even, but each piece must be aligned so that any bits can be used in the lower host portion. In this case, half of the block (a /17) is allocated to the Computer Science Dept, a quarter is allocated to the Electrical Engineering Dept. (a /18), and one eighth (a /19) to the Art Dept. The remaining eighth is unallocated.

A different way to see how the block was divided is to look at the resulting prefixes when written in binary notation:

Computer Science:	10000000	11010000	1 xxxxxxx	xxxxxxx
Electrical Eng.:	10000000	11010000	00 xxxxxx	xxxxxxx
Art:	10000000	11010000	011 xxxxx	xxxxxxx

Here, the vertical bar (|) shows the boundary between the subnet number and the host portion.



Splitting an IP prefix into separate networks with subnetting.

CIDR (Classless Inter- Domain Routing)

Splitting an address block into subnets, here we combine multiple small prefixes into a single larger prefix. This process is called **route aggregation**. The resulting larger prefix is sometimes called a **supernet**, to contrast with subnets as the division of blocks of addresses.

With aggregation, IP addresses are contained in prefixes of varying sizes. The same IP address that one router treats as part of a /22 (a block containing 210 addresses) may be treated by another router as part of a larger /20 (which contains 212 addresses). It is up to each router to have the corresponding prefix information. This design works with subnetting and is called **CIDR (Classless Inter- Domain Routing)**.

Let us consider an example in which a block of 8192 IP addresses is available starting at 194.24.0.0. Suppose that Cambridge University needs 2048 addresses and is assigned the addresses 194.24.0.0 through 194.24.7.255, along with mask 255.255.248.0. This is a /21 prefix.

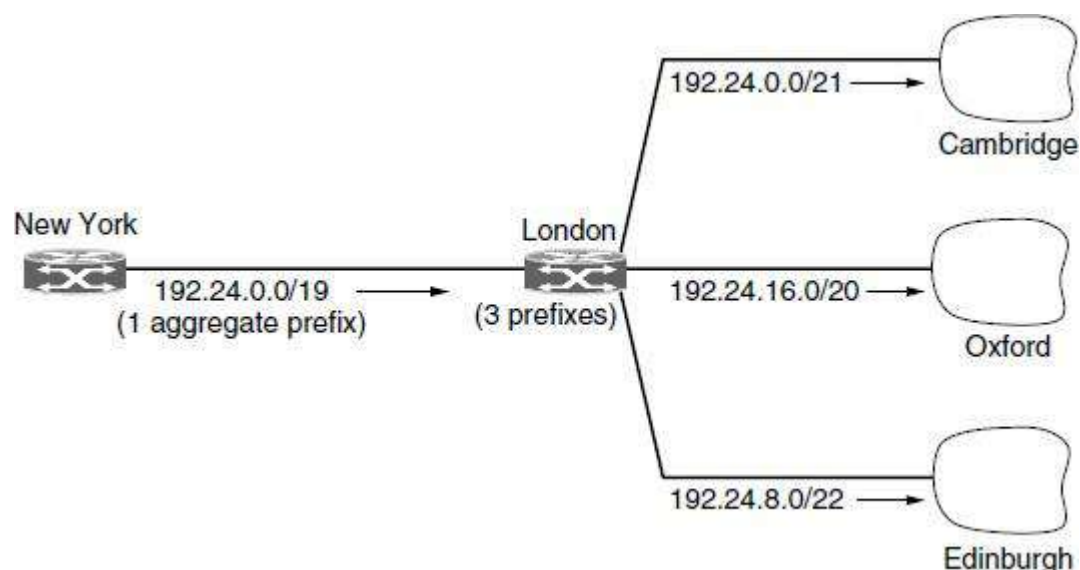
Next, Oxford University asks for 4096 addresses. Since a block of 4096 addresses must lie on a 4096-byte boundary, Oxford cannot be given addresses starting at 194.24.8.0. Instead, it gets 194.24.16.0 through 194.24.31.255, along with subnet mask 255.255.240.0.

Finally, the University of Edinburgh asks for 1024 addresses and is assigned addresses 194.24.8.0 through 194.24.11.255 and mask 255.255.252.0.

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

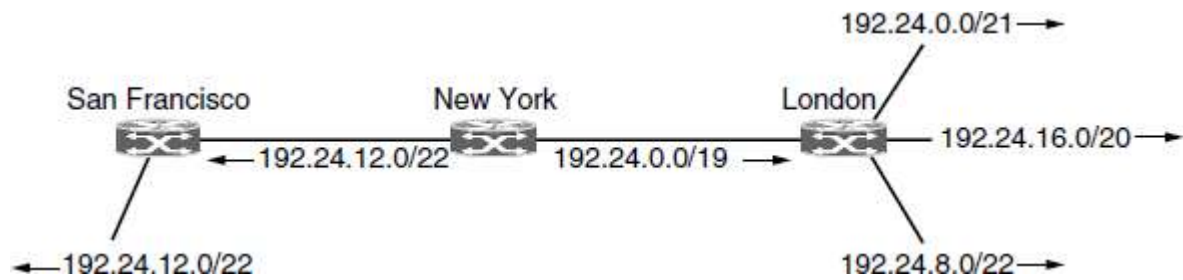
A set of IP address assignments

All of the routers in the default-free zone are now told about the IP addresses in the three networks. Routers close to the universities may need to send on a different outgoing line for each of the prefixes, so they need an entry for each of the prefixes in their routing tables. An example is the router in London in Figure.



Aggregation of IP prefixes

The rule is that packets are sent in the direction of the most specific route, or the **longest matching prefix** that has the fewest IP addresses. Longest matching prefix routing provides a useful degree of flexibility, as seen in the behavior of the router at New York in Figure.

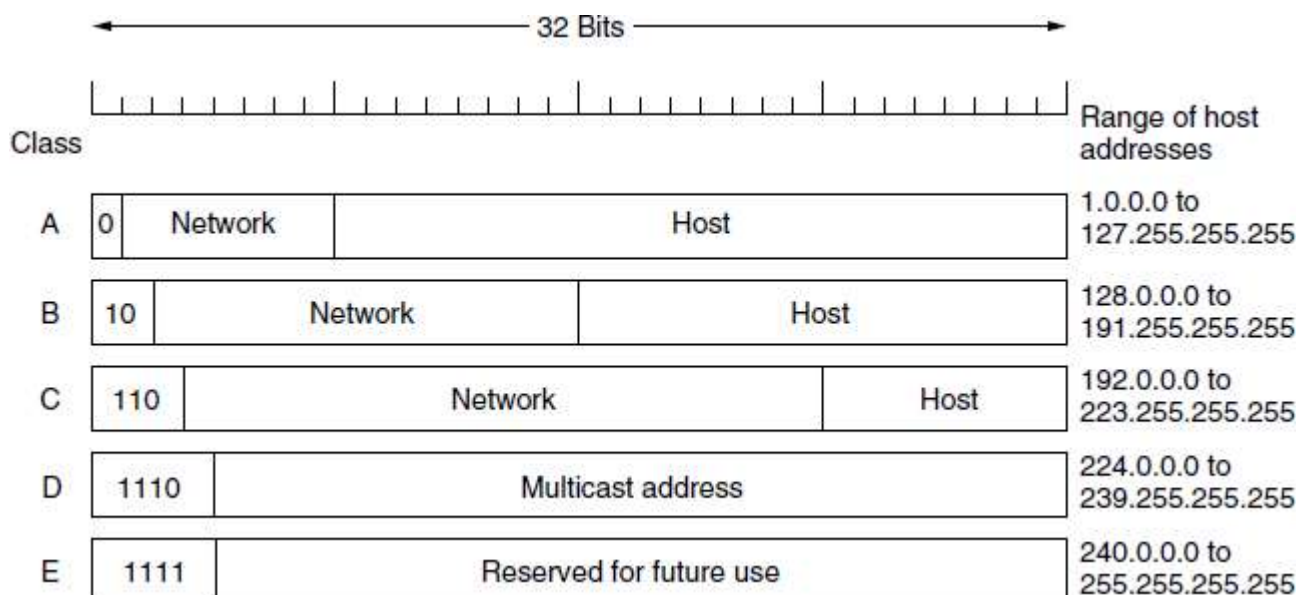


Longest matching prefix routing at the New York router

Classful and Special Addressing

IP addresses were divided into the five categories listed in Figure. This allocation has come to be called **classful addressing**.

- The class A, B, and C formats allow for up to 128 networks with 16 million hosts each, 16,384 networks with up to 65,536 hosts each, and 2 million networks (e.g., LANs) with up to 256 hosts each.
- Also supported is multicast (the class D format), in which a datagram is directed to multiple hosts.
- Addresses beginning with 1111 are reserved for use in the future.
- Class D addresses continue to be used in the Internet for multicast.



IP address formats

When a packet arrived at a router, a copy of the IP address was shifted right 28 bits to yield a 4-bit class number. A 16-way branch then sorted packets into A, B, C (and D and E) classes, with eight of the cases for class A, four of the cases for class B, and two of the cases for class C. The code for each class then

masked off the 8-, 16-, or 24-bit network number and right aligned it in a 32-bit word. The network number was then looked up in the A, B, or C table, usually by indexing for A and B networks and hashing for C networks. Once the entry was found, the outgoing line could be looked up and the packet forwarded.

Class D addresses continue to be used in the Internet for multicast. Actually, it might be more accurate to say that they are starting to be used for multicast.

There are also several other addresses that have special meanings.

0 0																																This host								
0 0								...								0 0								Host																A host on this network
1 1																																Broadcast on the local network								
Network																1 1 1 1								...								1 1 1 1								Broadcast on a distant network
127								(Anything)																								Loopback								

Special IP addresses

The IP address 0.0.0.0, the lowest address, is used by hosts when they are being booted. It means “this network” or “this host.” IP addresses with 0 as the network number refer to the current network. These addresses allow machines to refer to their own network without knowing its number.

The address consisting of all 1s, or 255.255.255.255—the highest address—is used to mean all hosts on the indicated network. It allows broadcasting on the local network, typically a LAN.

Finally, all addresses of the form 127.xx.yy.zz are reserved for loopback testing. Packets sent to that address are not put out onto the wire; they are processed locally and treated as incoming packets. This allows packets to be sent to the host without the sender knowing its number, which is useful for testing.

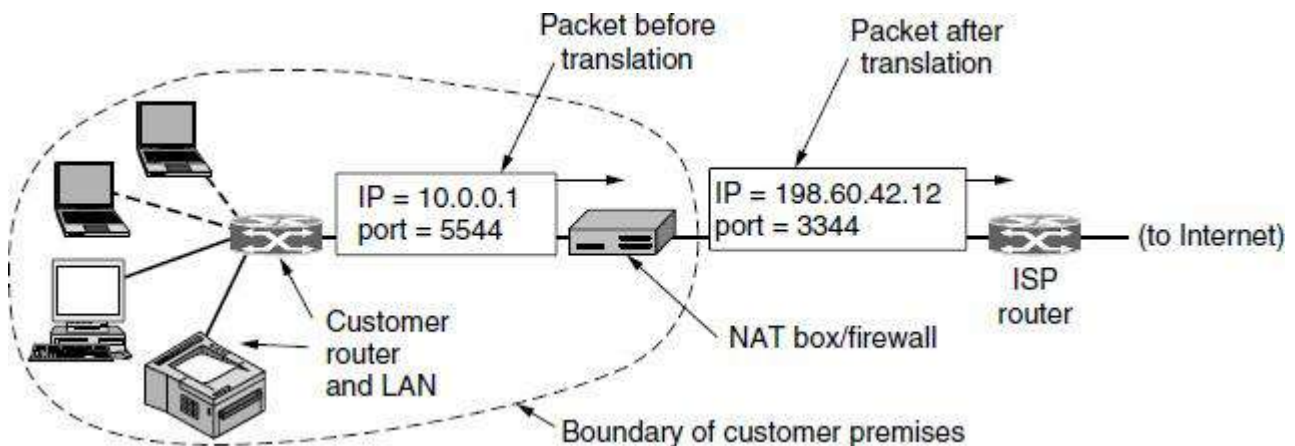
NAT—Network Address Translation

- The **NAT (Network Address Translation)**, is for the ISP to assign each home or business a single IP address (or at most, a small number of them) for Internet traffic. Within the customer network, every computer gets a unique IP address, which is used for routing intramural traffic.
- However, just before a packet exits the customer network and goes to the ISP, an address translation from the unique internal IP address to the shared public IP address takes place.
- This translation makes use of three ranges of IP addresses that have been declared as private. Networks may use them internally as they wish. The only rule is that no packets containing these addresses may appear on the Internet itself.

The three reserved ranges are:

10.0.0.0	-	10.255.255.255 / 8	(16,777,216 hosts)
172.16.0.0	-	172.31.255.255 / 12	(1,048,576 hosts)
192.168.0.0	-	192.168.255.255 / 16	(65,536 hosts)

The operation of NAT is shown in Figure. Every machine has a unique address of the form 10.x.y.z. However, before a packet leaves the customer premises, it passes through a NAT box that converts the internal IP source address, 10.0.0.1 in the figure, to the customer's true IP address, 198.60.42.12 in this example. The NAT box is often combined in a single device with a firewall, which provides security by carefully controlling what goes into the customer network and what comes out of it.



Placement and operation of a NAT box

When a process wants to establish a TCP connection with a remote process, it attaches itself to an unused TCP port on its own machine. This is called the **source port** and tells the TCP code where to send incoming packets belonging to this connection. The process also supplies a **destination port** to tell who to give the packets to on the remote side. Ports 0–1023 are reserved for well-known services.

19. Write a short notes on IPv6? (6 MARKS)

(NOV 2016)

- IPv6 (IP version 6) is a replacement design that does just that. It uses 128-bit addresses; a shortage of these addresses is not likely any time in the foreseeable future.
- However, IPv6 has proved very difficult to deploy. It is a different network layer protocol that does not really interwork with IPv4, despite many similarities.
- **SIPP (Simple Internet Protocol Plus)** was selected and given the designation IPv6.

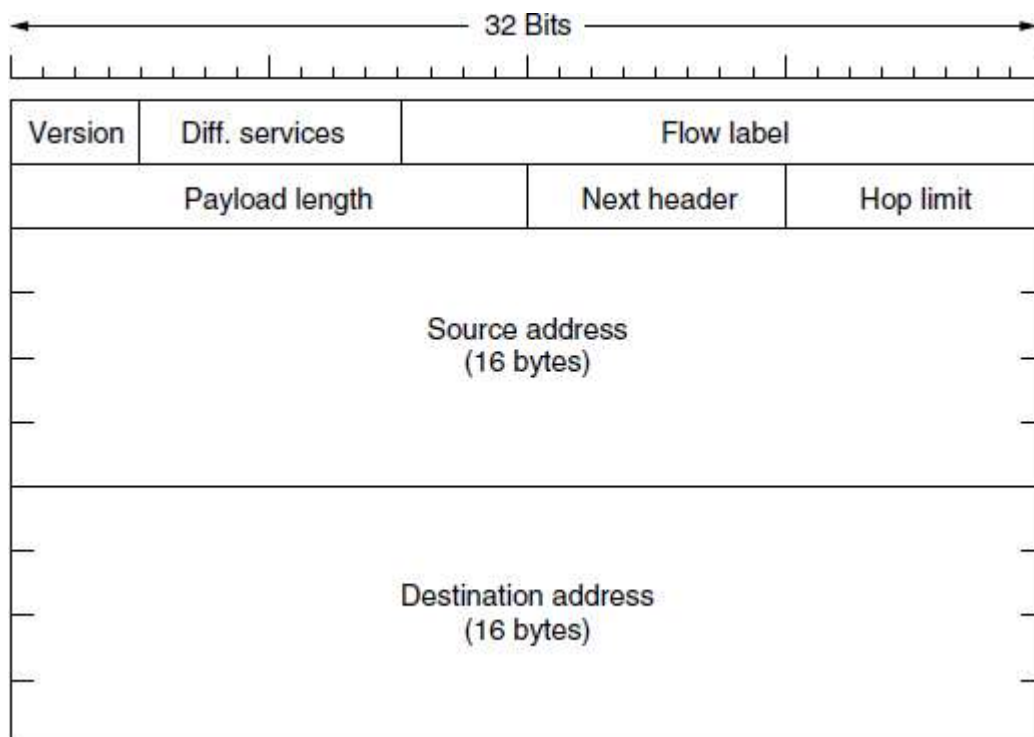
The main features of IPv6 are

1. First and foremost, IPv6 has longer addresses than IPv4. They are 128 bits long, which solves the problem that IPv6 set out to solve: providing an effectively unlimited supply of Internet addresses.
2. The second major improvement of IPv6 is the simplification of the header. It contains only seven fields (versus 13 in IPv4). This change allows routers to process packets faster and thus improves throughput and delay.

3. The third major improvement is better support for options. This change was essential with the new header because fields that previously were required are now optional.
4. A fourth area in which IPv6 represents a big advance is in security. Authentication and privacy are key features of the new IP.
5. Finally, more attention has been paid to quality of service.

The Main IPv6 Header

- The **Version field** is always 6 for IPv6 (and 4 for IPv4). For example, the Ethernet Type field has different values to indicate an IPv4 or an IPv6 payload.
- The **Differentiated services field** (originally called Traffic class) is used to distinguish the class of service for packets with different real-time delivery requirements.
- The **Flow label field** provides a way for a source and destination to mark groups of packets that have the same requirements and should be treated in the same way by the network.
- The **Payload length field** tells how many bytes follow the 40-byte header. The IPv4 **Total length field** because the meaning was changed slightly: the 40 header bytes are no longer counted as part of the length.
- The **Next header field** tells which transport protocol handler (e.g., TCP, UDP) to pass the packet to.
- The **Hop limit field** is used to keep packets from living forever.
- The Source address and Destination address fields occupy 16 bytes.



The IPv6 fixed header (required)

Extension Headers

Extension headers can be supplied to provide extra information, but encoded in an efficient way.

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

IPv6 extension headers

- Some of the headers have a fixed format; others contain a variable number of variable-length options.
- For these, each item is encoded as a **(Type, Length, Value)** tuple.
- The **Type** is a 1-byte field telling which option this is. The Type values have been chosen so that the first 2 bits tell routers that do not know how to process the option what to do.
- The choices are: skip the option; discard the packet; discard the packet and send back an ICMP packet; and discard the packet but do not send ICMP packets for multicast addresses.
- The **Length** is also a 1-byte field. It tells how long the value is (0 to 255 bytes).
- The **Value** is any information required, up to 255 bytes.

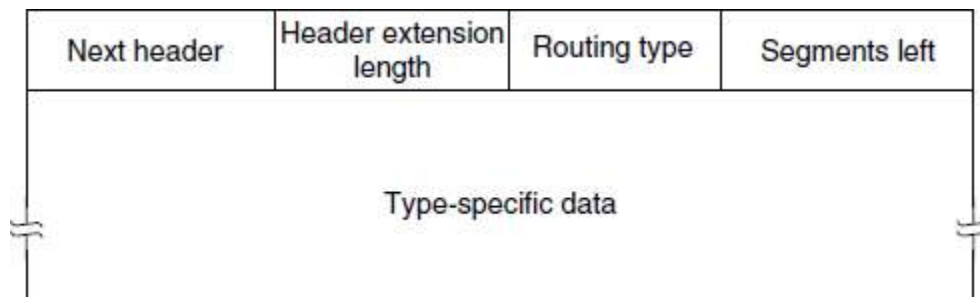
The hop-by-hop header is used for information that all routers along the path must examine. So far, one option has been defined: support of datagrams exceeding 64 KB. The format of this header is shown in Figure. When it is used, the Payload length field in the fixed header is set to 0.

Next header	0	194	4
Jumbo payload length			

The hop-by-hop extension header for large datagrams (jumbograms).

Datagrams using this header extension are called **jumbograms**. The use of jumbograms is important for supercomputer applications that must transfer gigabytes of data efficiently across the Internet.

The routing header lists one or more routers that must be visited on the way to the destination. It is very similar to the IPv4 loose source routing in that all addresses listed must be visited in order, but other routers not listed may be visited in between.



The extension header for routing

- The first 4 bytes of the routing extension header contain four 1-byte integers.
- The **Routing type field** gives the format of the rest of the header. Type 0 says that a reserved 32-bit word follows the first word, followed by some number of IPv6 addresses.
- Finally, the **Segments left field** keeps track of how many of the addresses in the list have not yet been visited. It is decremented every time one is visited.

20. Explain in detail about Internet Control Protocol? (11 MARKS)

- IP, which is used for data transfer, the Internet has several companion control protocols that are used in the network layer.
- They include ICMP, ARP, and DHCP.

ICMP (Internet Control Message Protocol)

- The operation of the Internet is monitored closely by the routers.
- When something unexpected occurs during packet processing at a router, the event is reported to the sender by the **ICMP (Internet Control Message Protocol)**.
- ICMP is also used to test the Internet.
- Each ICMP message type is carried encapsulated in an IP packet.

The principal ICMP message types

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

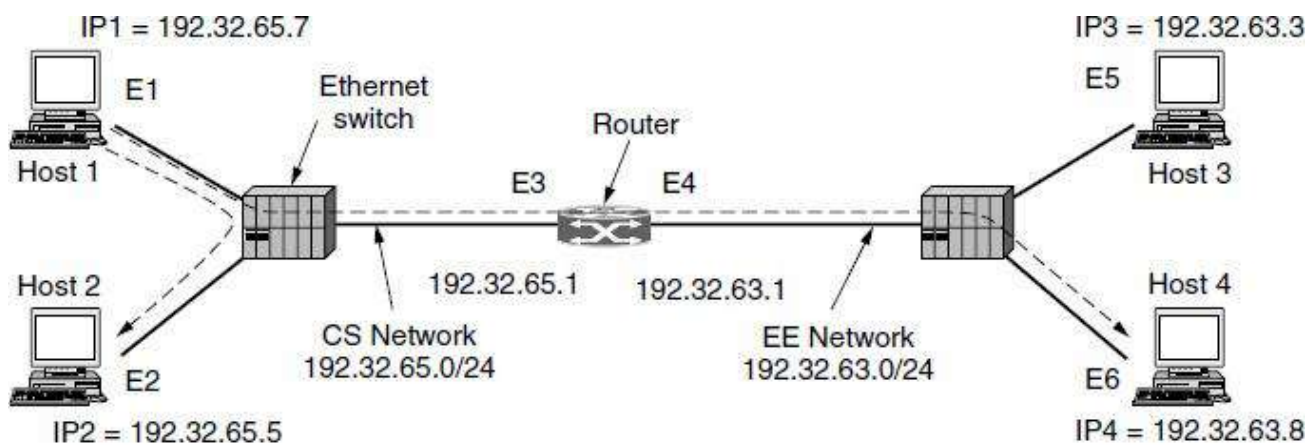
- The DESTINATION UNREACHABLE message is used when the router cannot locate the destination or when a packet with the DF bit cannot be delivered because a “small-packet” network stands in the way.
- The TIME EXCEEDED message is sent when a packet is dropped because its TtL (Time to live) counter has reached zero. This event is a symptom that packets are looping, or that the counter values are being set too low.
- The PARAMETER PROBLEM message indicates that an illegal value has been detected in a header field. This problem indicates a bug in the sending host’s IP software or possibly in the software of a router transited.
- The SOURCE QUENCH message was long ago used to throttle hosts that were sending too many packets. When a host received this message, it was expected to slow down.
- The REDIRECT message is used when a router notices that a packet seems to be routed incorrectly. It is used by the router to tell the sending host to update to a better route.
- The ECHO and ECHO REPLY messages are sent by hosts to see if a given destination is reachable and currently alive. Upon receiving the ECHO message, the destination is expected to send back an ECHO REPLY message. These messages are used in the ping utility that checks if a host is up and on the Internet.
- The TIMESTAMP REQUEST and TIMESTAMP REPLY messages are similar, except that the arrival time of the message and the departure time of the reply are recorded in the reply. This facility can be used to measure network performance.
- The ROUTER ADVERTISEMENT and ROUTER SOLICITATION messages are used to let hosts find nearby routers. A host needs to learn the IP address of at least one router to be able to send packets off the local network.

ARP (The Address Resolution Protocol)

Let us use the example of Figure, in which a small university with two /24 networks is illustrated. One network (CS) is a switched Ethernet in the Computer Science Dept. It has the prefix 192.32.65.0/24. The other LAN (EE), also switched Ethernet, is in Electrical Engineering and has the prefix 192.32.63.0/24. The two LANs are connected by an IP router. Each machine on an Ethernet and each interface on the router has a unique Ethernet address, labeled E1 through E6, and a unique IP address on the CS or EE network.

- The protocol used for asking this question and getting the reply is called **ARP (Address Resolution Protocol)**. Almost every machine on the Internet runs it. ARP is defined in RFC 826.
- The advantage of using ARP over configuration files is the simplicity. The system manager does not have to do much except assign each machine an IP address and decide about subnet masks.

At this point, the IP software on host 1 builds an Ethernet frame addressed to E2, puts the IP packet (addressed to 192.32.65.5) in the payload field, and dumps it onto the Ethernet. The IP and Ethernet addresses of this packet are given in Figure. The Ethernet NIC of host 2 detects this frame, recognizes it as a frame for itself, scoops it up, and causes an interrupt. The Ethernet driver extracts the IP packet from the payload and passes it to the IP software, which sees that it is correctly addressed and processes it.



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

Two switched Ethernet LANs joined by a router

There should not be a response, but a side effect of the broadcast is to make or update an entry in everyone's ARP cache. This is known as a **gratuitous ARP**. If a response does (unexpectedly) arrive, two machines have been assigned the same IP address. The error must be resolved by the network manager before both machines can use the network.

The router will then receive frames sent to 192.32.63.8 and forward them onto the EE network. This solution is called **proxy ARP**. It is used in special cases in which a host wants to appear on a network even though it actually resides on another network.

A common situation, for example, is a mobile computer that wants some other node to pick up packets for it when it is not on its home network.

DHCP (The Dynamic Host Configuration Protocol)

- DHCP provides static and dynamic address allocation that can be manual or automatic.
- It is possible to manually configure each computer, but that is tedious and error-prone. There is a better way, and it is called **DHCP (Dynamic Host Configuration Protocol)**.

With DHCP, every network must have a DHCP server that is responsible for configuration. When a computer is started, it has a built-in Ethernet or other link layer address embedded in the NIC, but no IP address. Much like ARP, the computer broadcasts a request for an IP address on its network. It does this by using a DHCP DISCOVER packet. This packet must reach the DHCP server. If that server is not directly attached to the network, the router will be configured to receive DHCP broadcasts and relay them to the DHCP server, wherever it is located.

When the server receives the request, it allocates a free IP address and sends it to the host in a DHCP OFFER packet (which again may be relayed via the router). To be able to do this work even when hosts do not have IP addresses, the server identifies a host using its Ethernet address (which is carried in the DHCP DISCOVER packet).

An issue that arises with automatic assignment of IP addresses from a pool is for how long an IP address should be allocated. If a host leaves the network and does not return its IP address to the DHCP server, that address will be permanently lost. After a period of time, many addresses may be lost.

- It is widely used in the Internet to configure all sorts of parameters in addition to providing hosts with IP addresses.
- As well as in business and home networks, DHCP is used by ISPs to set the parameters of devices over the Internet access link, so that customers do not need to phone their ISPs to get this information.
- Common examples of the information that is configured include the network mask, the IP address of the default gateway, and the IP addresses of DNS and time servers.

21. Explain the following internet control protocols. (11 MARKS) (NOV 2015, 2016)

(a) OSPF (b) BGP

OSPF - An Interior Gateway Routing Protocol

- Internet is made up of a large number of independent networks or **ASes (Autonomous Systems)** that are operated by different organizations, usually a company, university, or ISP.
- Inside of its own network, an organization can use its own algorithm for **internal routing**, or **intradomain routing**.
- An intradomain routing protocol is also called an **interior gateway protocol**.
- The ARPANET switched over to a link state protocol in May 1979 because of these problems, and in 1988 IETF began work on a link state protocol for **intradomain routing**. That protocol, called **OSPF (Open Shortest Path First)**, became a standard in 1990.
- It drew on a protocol called **IS-IS (Intermediate-System to Intermediate-System)**, which became an ISO standard.
- The Open Shortest Path First or OSPF protocol is an intradomain routing protocol based on link state routing. Its domain is also an autonomous system.

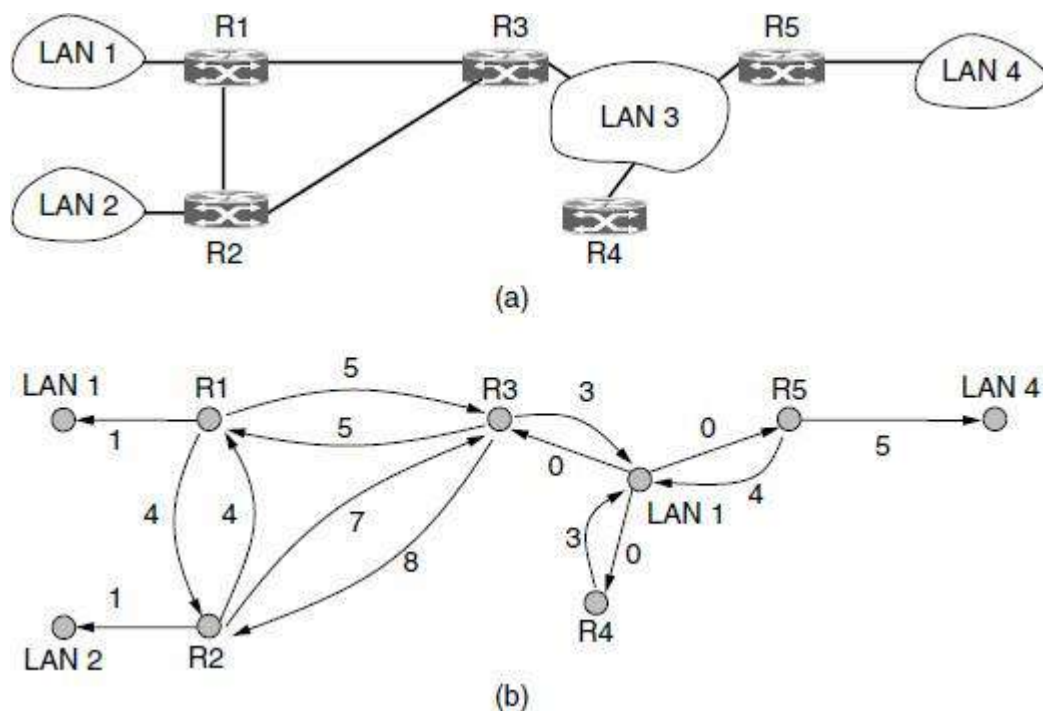
With other routing protocols, the group designing OSPF had a long list of requirements

1. First, the algorithm had to be published in the open literature, hence the “O” in OSPF.
2. Second, the new protocol had to support a variety of distance metrics, including physical distance, delay, and so on.
3. Third, it had to be a dynamic algorithm, one that adapted to changes in the topology automatically and quickly.
4. Fourth, and new for OSPF, it had to support routing based on type of service.

5. Fifth, and related to the above, OSPF had to do load balancing, splitting the load over multiple lines.
6. Sixth, support for hierarchical systems was needed.
7. Seventh, some modicum of security was required to prevent fun-loving students from spoofing routers by sending them false routing information.

OSPF supports both point-to-point links (e.g., SONET) and broadcast networks (e.g., most LANs). Actually, it is able to support networks with multiple routers, each of which can communicate directly with the others (called **multi-access networks**).

An example of an autonomous system network is given in Figure (a). Hosts are omitted because they do not generally play a role in OSPF, while routers and networks do. Most of the routers in Figure (a) are connected to other routers by point-to-point links, and to networks to reach the hosts on those networks. However, routers R3, R4, and R5 are connected by a broadcast LAN such as switched Ethernet.



(a) An autonomous system. (b) A graph representation of (a).

OSPF operates by abstracting the collection of actual networks, routers, and links into a directed graph in which each arc is assigned a weight (distance, delay, etc.). A point-to-point connection between two routers is represented by a pair of arcs, one in each direction. Their weights may be different. A broadcast network is represented by a node for the network itself, plus a node for each router. The arcs from that network node to the routers have weight 0.

Figure (b) shows the graph representation of the network of Figure (a). What OSPF fundamentally does is represent the actual network as a graph like this and then use the link state method to have every

router compute the shortest path from itself to all other nodes. Multiple paths may be found that are equally short. In this case, OSPF remembers the set of shortest paths and during packet forwarding, traffic is split across them. This helps to balance load. It is called **ECMP (Equal Cost MultiPath)**.

To work at this scale, OSPF allows an AS to be divided into numbered areas, where an area is a network or a set of contiguous networks. Routers that lie wholly within an area are called **internal routers**. An area is a generalization of an individual network. Outside an area, its destinations are visible but not its topology. This characteristic helps routing to scale.

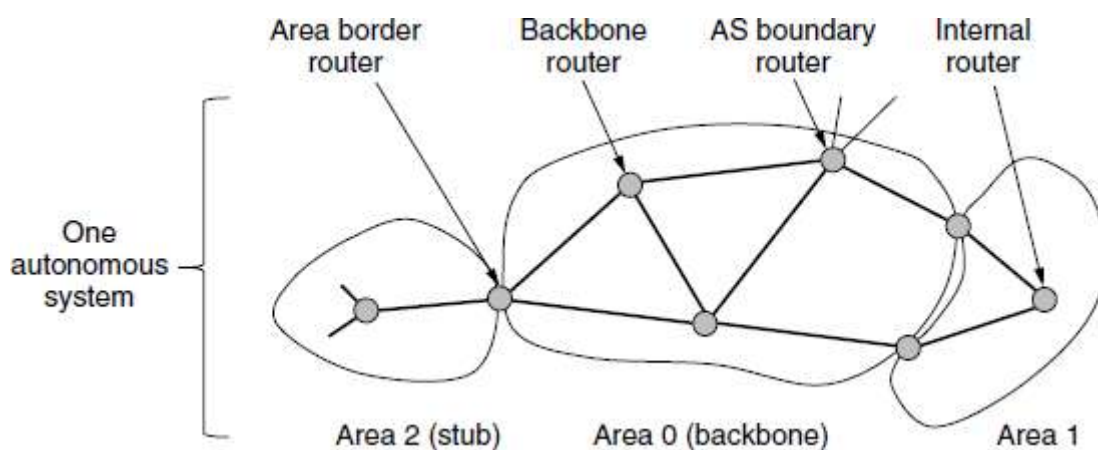
Every AS has a **backbone area**, called area 0. The routers in this area are called **backbone routers**. All areas are connected to the backbone, possibly by tunnels, so it is possible to go from any area in the AS to any other area in the AS via the backbone. A tunnel is represented in the graph as just another arc with a cost. As with other areas, the topology of the backbone is not visible outside the backbone.

Each router that is connected to two or more areas is called an **area border router**. It must also be part of the backbone. The job of an area border router is to summarize the destinations in one area and to inject this summary into the other areas to which it is connected. This summary includes cost information but not all the details of the topology within an area.

Routes to destinations out of the area always start with the instruction “Go to the border router.” This kind of area is called a **stub area**.

The last kind of router is the **AS boundary router**. It injects routes to external destinations on other ASes into the area. The external routes then appear as destinations that can be reached via the AS boundary router with some cost. An external route can be injected at one or more AS boundary routers.

The relationship between ASes, areas, and the various kinds of routers is shown in Figure. One router may play multiple roles, for example, a border router is also a backbone router.



The relation between ASes, backbones, and areas in OSPF

During normal operation, each router within an area has the same link state database and runs the same shortest path algorithm. Its main job is to calculate the shortest path from itself to every other router and network in the entire AS. An area border router needs the databases for all the areas to which it is connected and must run the shortest path algorithm for each area separately.

For a source and destination in the same area, the best intra-area route (that lies wholly within the area) is chosen. For a source and destination in different areas, the inter-area route must go from the source to the backbone, across the backbone to the destination area, and then to the destination. This algorithm forces a star configuration on OSPF, with the backbone being the hub and the other areas being spokes. Because the route with the lowest cost is chosen, routers in different parts of the network may use different area border routers to enter the backbone and destination area. Packets are routed from source to destination “as is.”

When a router boots, it sends HELLO messages on all of its point-to-point lines and multicasts them on LANs to the group consisting of all the other routers. From the responses, each router learns who its neighbors are. Routers on the same LAN are all neighbors.

OSPF works by exchanging information between adjacent routers, which is not the same as between neighboring routers. In particular, it is inefficient to have every router on a LAN talk to every other router on the LAN. To avoid this situation, one router is elected as the **designated router**. It is said to be **adjacent** to all the other routers on its LAN, and exchanges information with them. In effect, it is acting as the single node that represents the LAN. Neighboring routers that are not adjacent do not exchange information with each other. A backup designated router is always kept up to date to ease the transition should the primary designated router crash and need to be replaced immediately.

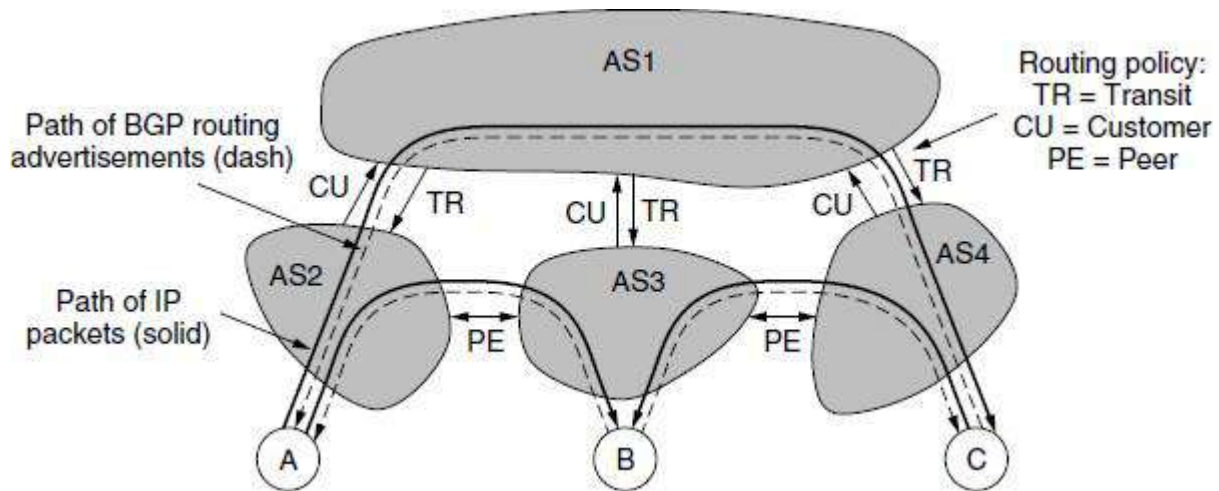
The five kinds of messages are

Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

BGP—The Exterior Gateway Routing Protocol

- Border Gateway Protocol (BGP) is an interdomain routing protocol using path vector routing.
- One common policy is that a customer ISP pays another provider ISP to deliver packets to any other destination on the Internet and receive packets sent from any other destination. The customer ISP is said to buy **transit service** from the provider ISP. This is just like a customer at home buying Internet access service from an ISP.

An example of transit service



Routing policies between four autonomous systems

There are four ASes that are connected. The connection is often made with a link at **IXPs (Internet eXchange Points)**, facilities to which many ISPs have a link for the purpose of connecting with other ISPs. AS2, AS3, and AS4 are customers of AS1. They buy transit service from it. Thus, when source A sends to destination C, the packets travel from AS2 to AS1 and finally to AS4. The routing advertisements travel in the opposite direction to the packets. AS4 advertises C as a destination to its transit provider, AS1, to let sources reach C via AS1. Later, AS1 advertises a route to C to its other customers, including AS2, to let the customers know that they can send traffic to C via AS1.

Types of Autonomous Systems

The Internet is divided into hierarchical domains called autonomous systems. For example, a large corporation that manages its own network and has full control over it is an autonomous system. A local ISP that provides services to local customers is an autonomous system.

We can divide autonomous systems into three categories:

1. stub,
2. multi-homing, and
3. transit

Stub AS

However, it does not need to run BGP because it is a **stub** network that is connected to the rest of the Internet by only one link. So the only place for it to send packets destined outside of the network is over the link to AS2. There is nowhere else to go. This path can be arranged simply by setting up a default route. A good example of a stub AS is a small corporation or a small local ISP.

Multi-homing

On the other hand, some company networks are connected to multiple ISPs. This technique is used to improve reliability, since if the path through one ISP fails, the company can use the path via the other ISP.

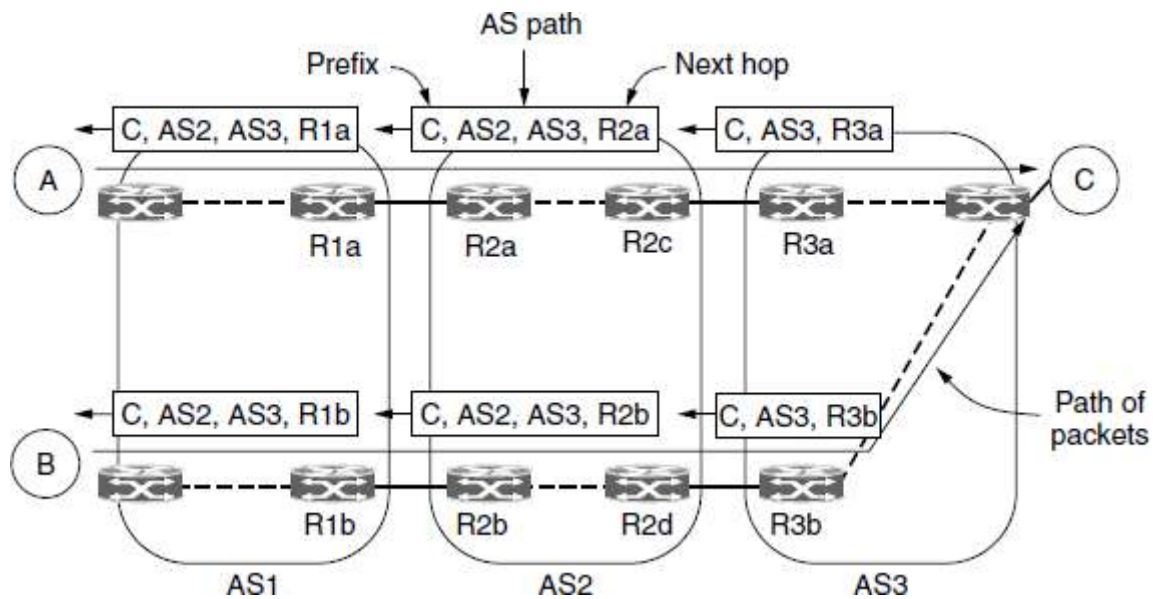
This technique is called **multihoming**. A good example of a multi-homed AS is a large corporation that is connected to more than one regional or national AS that does not allow transient traffic.

Transit AS

A transit AS is a multi-homed AS that also allows transient traffic. Good examples of transit ASs are national and international ISPs (Internet backbones).

- BGP is a form of distance vector protocol, but it is quite unlike intradomain distance vector protocols such as RIP.
- Another large difference is that instead of maintaining just the cost of the route to each destination, each BGP router keeps track of the path used. This approach is called a **path vector protocol**.
- The path consists of the next hop router (which may be on the other side of the ISP, not adjacent) and the sequence of ASes, or **AS path**, that the route has followed (given in reverse order).
- Finally, pairs of BGP routers communicate with each other by establishing TCP connections. Operating this way provides reliable communication and also hides all the details of the network being passed through.

An example of how BGP routes are advertised is shown



Propagation of BGP route advertisements

There are three ASes and the middle one is providing transit to the left and right ISPs. A route advertisement to prefix C starts in AS3. When it is propagated across the link to R2c at the top of the figure, it has the AS path of simply AS3 and the next hop router of R3a. At the bottom, it has the same AS path but a different next hop because it came across a different link. This advertisement continues to propagate and crosses the boundary into AS1. At router R1a, at the top of the figure, the AS path is AS2, AS3 and the next hop is R2a.

To propagate BGP routes from one side of the ISP to the other, so they can be sent on to the next ISP. This task could be handled by the intradomain protocol, but because BGP is very good at scaling to large networks, a variant of BGP is often used. It is called **iBGP (internal BGP)** to distinguish it from the regular use of BGP as **eBGP (external BGP)**.

PONDICHERRY UNIVERSITY QUESTIONS

2 MARKS

1. Bring out the difference between IPv4 and IPv6 address. **(NOV 2015) (Ref.Qn.No.60, Pg.no.11)**
2. Why is admission control required in networks? **(NOV 2015) (Ref.Qn.No.42, Pg.no.8)**
3. Why traffic shaping approach is used in congestion. **(MAY 2016) (Ref.Qn.No.43, Pg.no.8)**
4. State the difference between classless and classful addressing. **(NOV 2016) (Ref.Qn.No.58, Pg.no.10)**
5. Find the error, if any in the following IP addresses. **(NOV 2016) (Ref.Qn.No.69, Pg.no.12)**
(a) 111.56.045.78 (b) 75.45.301.14
6. State the purpose of maintaining the routing table and where is it maintained. **(MAY 2017) (Ref.Qn.No.19, Pg.no.5)**

11 MARKS

NOV 2015 (REGULAR)

1. (a). Discuss the optimality principle of routing. (6) **(Ref.Qn.No.3, Pg.no.19)**
(b). Write note on distance vector routing. (5) **(Ref.Qn.No.6, Pg.no.25)**
(OR)
2. (a). What are the approaches for congestion control in networks? (6) **(Ref.Qn.No.12, Pg.no.37)**
(b). Write notes on OSPF and BGP protocols. (5) **(Ref.Qn.No.21, Pg.no.65)**

MAY 2016 (ARREAR)

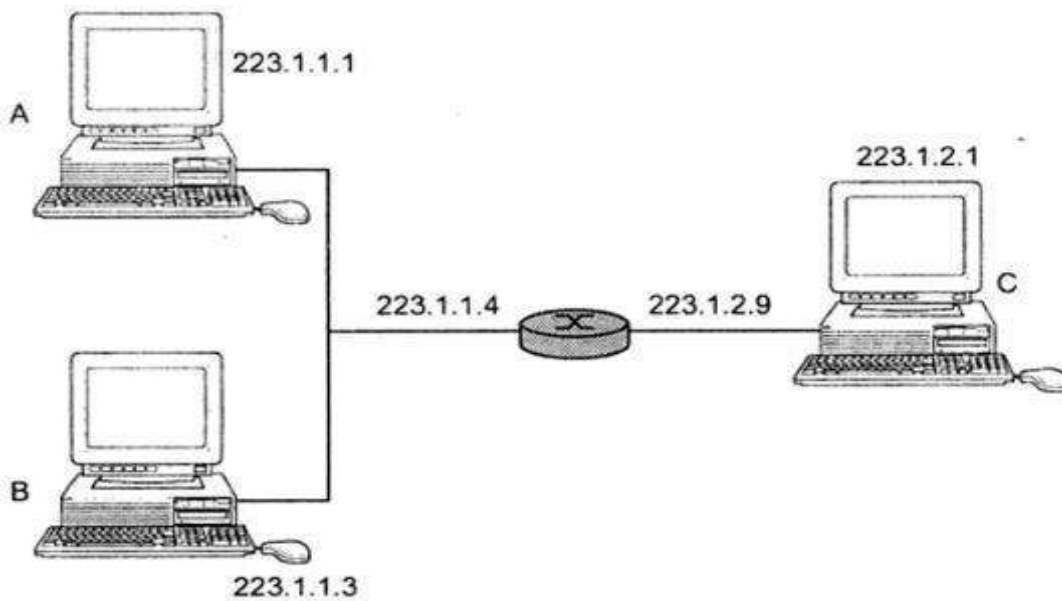
1. (a). Explain the services provided to the transport layer. (7) **(Ref.Qn.No.1, Pg.no.13)**
(b). What is fragmentation? Explain briefly. (4) **(Ref.Qn.No.16, Pg.no.48)**
(OR)
2. Explain briefly about Internetworking and also explain the networks differ. **(Ref.Qn.No.13, Pg.no.43)**

NOV 2016 (REGULAR)

1. (a). Explain in detail about IPV6 and compare with IPV4. **(Ref.Qn.No.17, 19, Pg.no.50,59)**
(b). State the major difference between Vector Routing and Link State Routing. **(Ref.Qn.No.6, 7, Pg.no.25, 28)**
(OR)
2. Explain the following internet control protocols. **(Ref.Qn.No.21, Pg.no.65)**
(a) OSPF (b) BGP

MAY 2017 (ARREAR)

1. Consider the following network with three hosts and one router. The IP addresses of the different interfaces in the network are indicated on the figure. Assume that the MAC address of node A is 00: 02: 98: FF: CD: 00 and that its ARP table is initially empty. Given that B MAC address is 00: 02: 98: FF: CD: 02, describe all the steps involved if A wants to send an IP datagram to B and to show the headers of all frames transmitted.



(OR)

2. Explain the working of Distance Vector Routing. **(Ref.Qn.No.6, Pg.no.25)**